

they are waveshaping circuits.

* Clippers :

Clippers are networks that use diodes to clip a portion of input signal without distorting the remaining part of the waveform.

Two types:

1. Series → diode is connected in series with load resistance
2. Parallel → diode is connected in parallel with load resistance.



⇒ Series Clipper

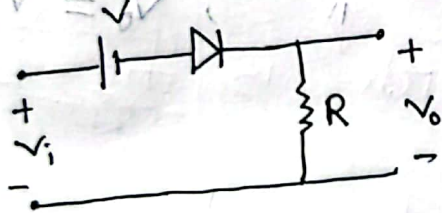
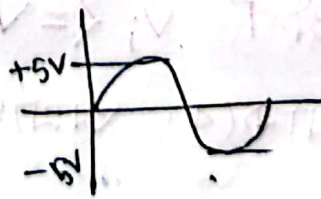
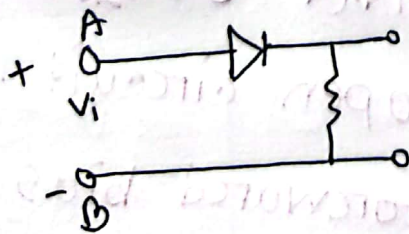


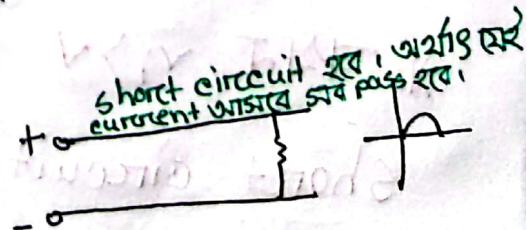
Fig: Series clipper with a dc supply.

Example :



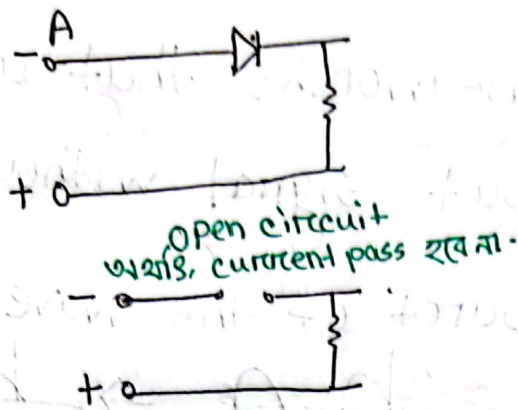
1st half cycle:

$$\begin{aligned} A &= + \\ B &= - \end{aligned}$$



Diode → forward bias

2nd half cycle:



$$A = -$$

$$B = +$$

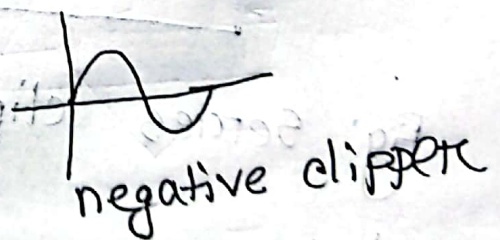
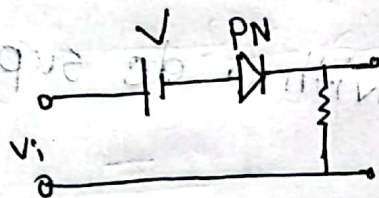
Diode $\Rightarrow$ reverse biased

এখানে negative part pass হচ্ছে না. অর্থাৎ এটা series negative clipper. কারণ negative কে clipped করছে।

$\Rightarrow$ when diode is open circuit, $V_i = V$

$\Rightarrow$ " " " short " " , $V_o = V_i - V$

Example:



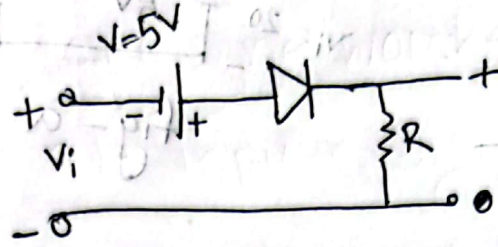
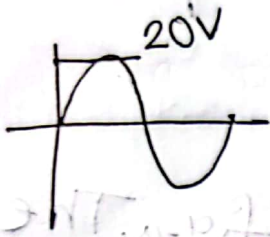
* যখন $V_i \leq V$ থাকবে তখন reverse bias থাকবে. অর্থাৎ, open circuit. $V_o = 0$

* যখন $V_i > V$ হবে তখন forward bias হবে. অর্থাৎ, short circuit হবে. $V_o = V_i - V$

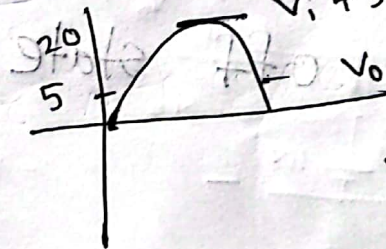
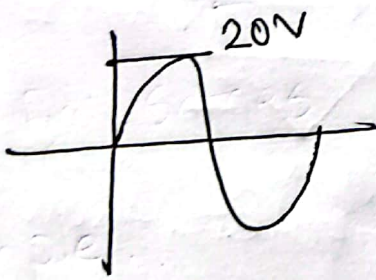
*Example = 2.20:-

Determine the output for the

network:



Here, $V_o = V_i + 5V$. Substituting $i_d = 0$ at $V_d = 0$ for transition level we obtain $V_i = -5V$ for V_i more negative than $-5V$ the diode will enter its open circuit state, while for voltage more positive than $-5V$ the diode is in short-circuit state.

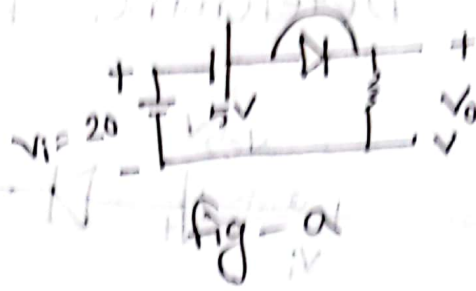
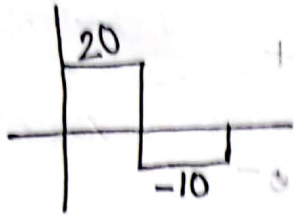


$$V_i + 5V = 20V + 5V = 25V$$

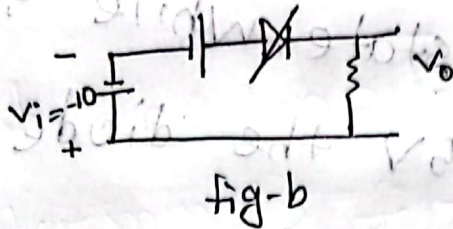
$$V_o = 0 + 5V = 5V$$

$$V_o = -5V + 5V = 0$$

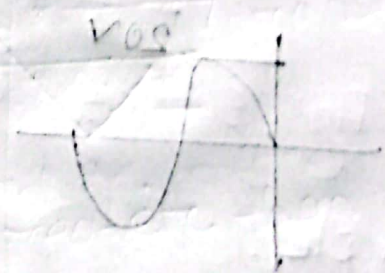
*Example = 2.21: Determine the output for the network.



For $v_i = 20V$ the network is Fig-a. The diode is in the short circuit state and $v_o = 20 + 5 = 25V$



For $v_i = -10V$ the network is Fig-b, placing the diode in off state

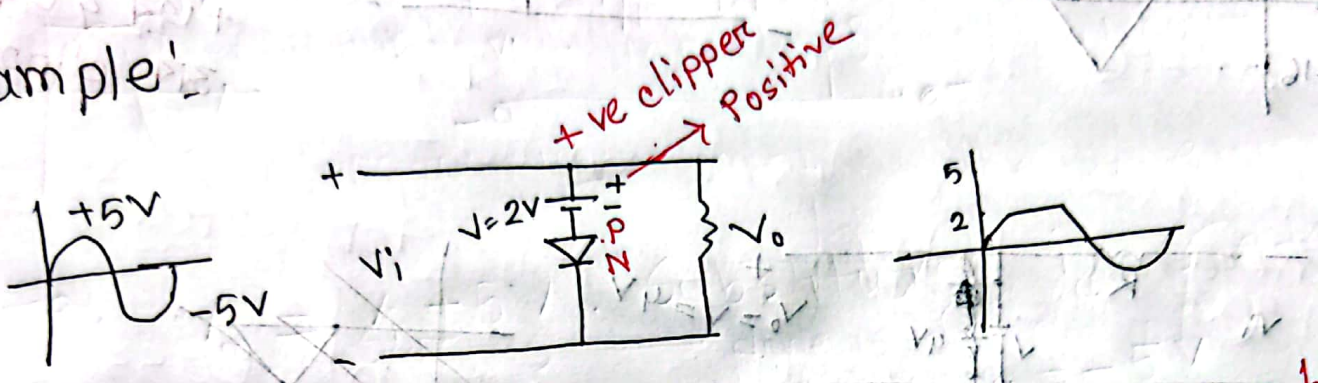


⇒ Parallel Clipper

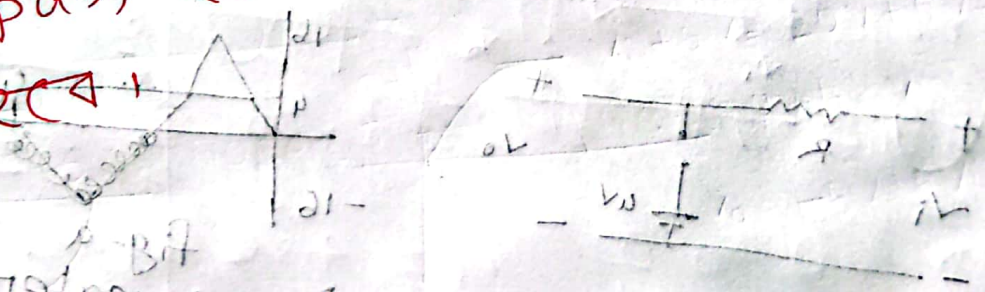
Here diode is set parallel

with circuit.

Example

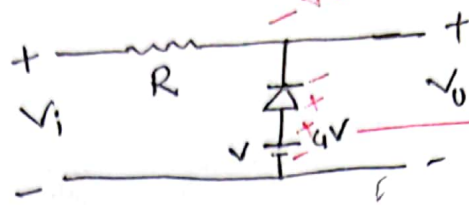
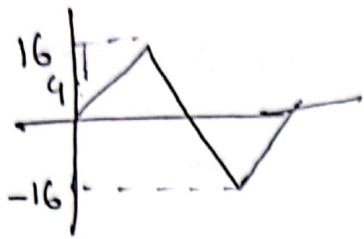


*Parallel এ $V_i \leq V$ থাকলে diode reverse bias থাকবে। Parallel থাকায় current যা আসবে তাই pass হবে। অর্থাৎ, short circuit. diode এ current হবে না। $V_i > V$ হলে forward bias হবে। অর্থাৎ current diode এ pass হবে। V_o ত আর যাবে না, open circuit হবে।

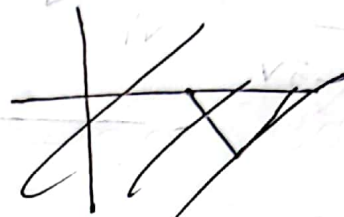
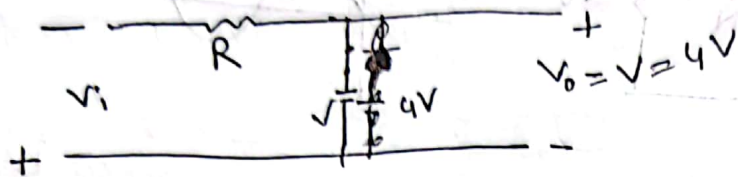


When the input voltage is less than the diode's forward voltage, the diode is in reverse bias and no current flows. When the input voltage exceeds the forward voltage, the diode is in forward bias and current flows, clipping the output at the forward voltage level.

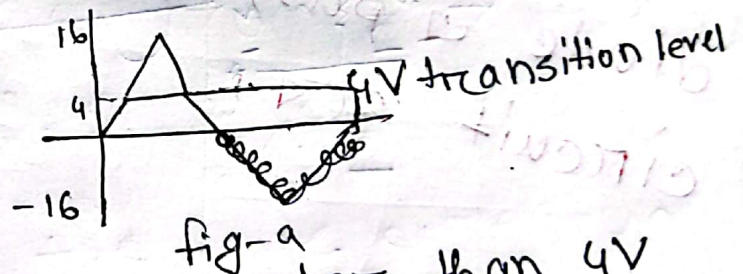
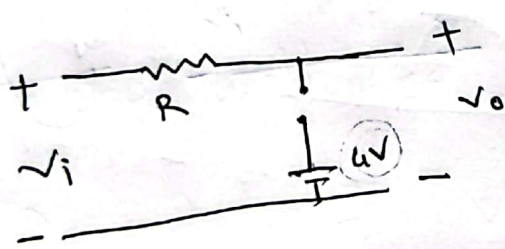
* Example - 2.22 : Determine v_o for the network



-ve clipper
+4V, তাই
+এর 4 এর উপর
অব clipped হবে

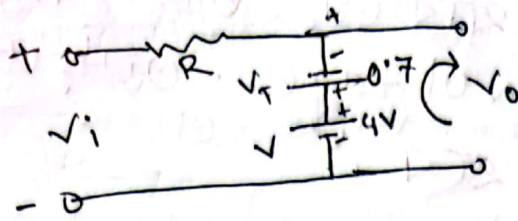


The diode will be "on" state for the negative region of the input signal, where the defined terminals for v_o require that $v_o = v = 4V$



when the input voltage is greater than 4V the diode will be in "off" state. Any input less than 4V will result in a short-circuited diode. So final v_o is fig-a. Completing the sketch of v_o result in waveform is fig-a.

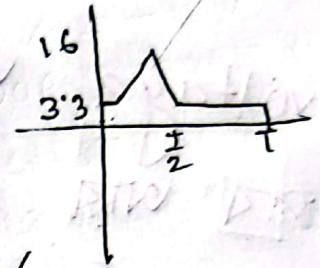
* Example = 2.23: Determine v_o for the network using a silicon diode with $V_T = 0.7V$.



Applying Kirchhoff's ~~law~~ voltage law around the output loop in the ~~clockwise~~ clockwise direction we find that:

$$-V + V_T + v_i = 0$$

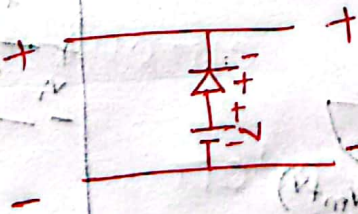
$$\text{and, } v_i = V - V_T = 4 - 0.7 = 3.3V$$



For input voltage greater than 3.3V the diode will be an open circuit and $v_o = v_i$. For input voltage of less than 3.3V, the diode will be

in on state.

* Parallel $\wedge$ diode $\wedge$ battery যদি এমন থাকে



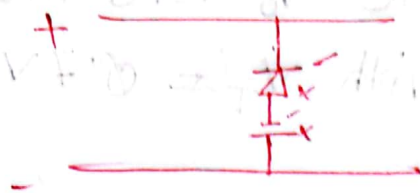
v - voltage পর্যন্ত
input voltage diode
হয়ে pass হবে।

negative half cycle $\wedge$ v

এ নিচে যত যাব তত direct
বাবার আইন দেয়া হয়েছে।

pass হবে তাই v voltage

*



Positive half cycle এ
দুইটি reverse এ
শাকবে ওই input
voltage = output voltage
হবে।

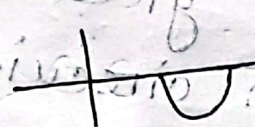
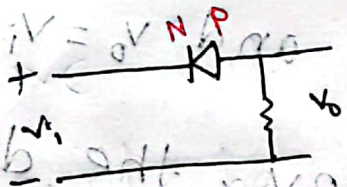
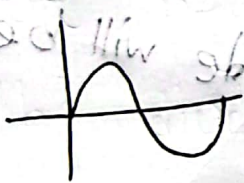
negative half cycle এ forward এ থাকে

v voltage অনুযায়ী current pass হবে।

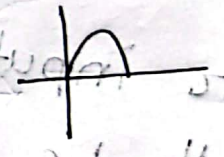
এরপর আর হবে না।

* Simple series = clipper (Ideal Diode):

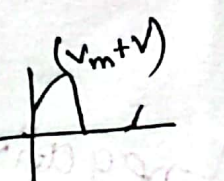
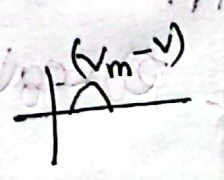
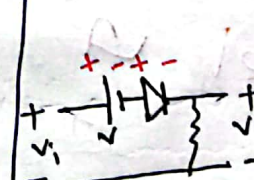
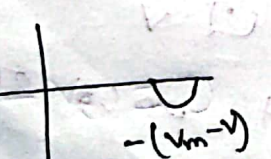
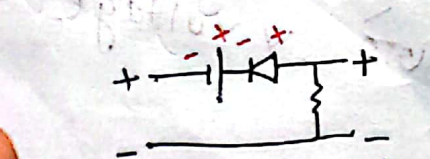
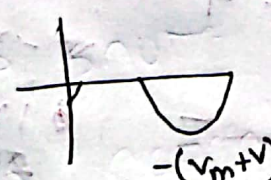
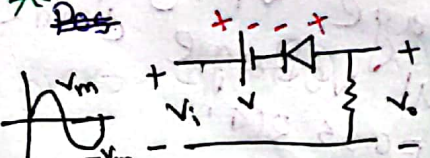
Positive



Negative

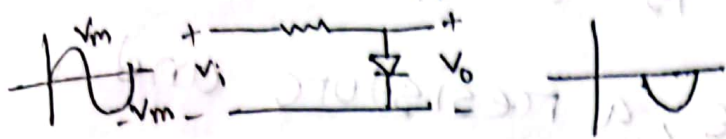


* Biased Series clipper

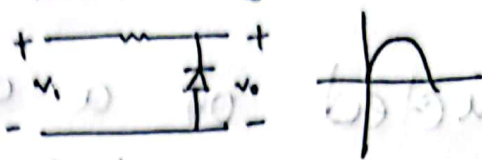


* Simple Parallel Clippers!

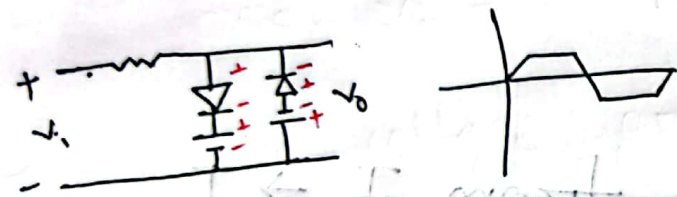
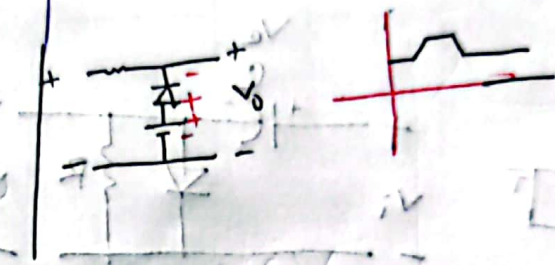
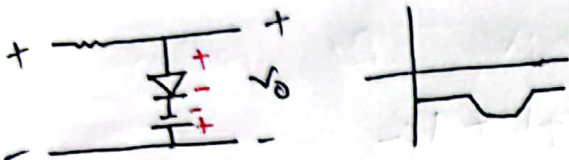
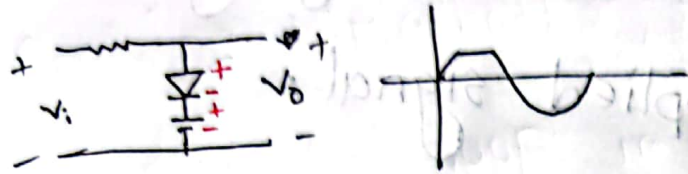
Positive



Negative

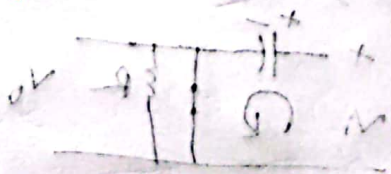


* Biased Parallel



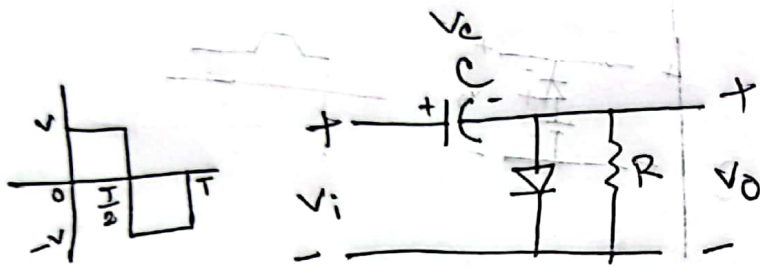
* Parallel এ diode এর negative এবং battery এর negative পরস্পর যুক্ত থাকে তবে অব voltage negative waveform এ আসবে।

* আর diode এর positive এবং battery এর positive যুক্ত থাকে তবে voltage positive waveform এ আসবে।



* Clamper:

A clamper is a network constructed of a diode, a resistor and a capacitor that shifts the wave form to a different dc level without changing the appearance of the applied signal.



Time constant, $\tau = RC$

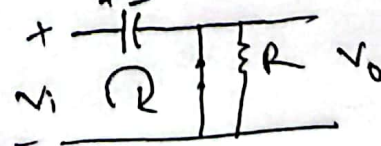
For negative half cycle from $\frac{T}{2} \rightarrow T$ the diode will be off. So, $\tau \gg \frac{T}{2}$

From 0 to $\frac{T}{2}$ $v_i = +V$, the direction is clockwise, and diode is forward biased. So it is short circuit. So, $v_o = 0$

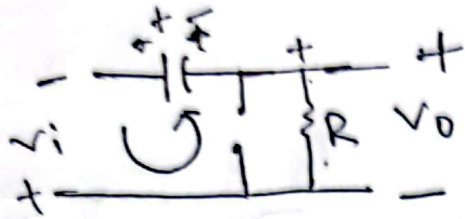
voltage across capacitor, v_c

$$v_i - v_c = 0$$

$$v_c = v_i = +V$$



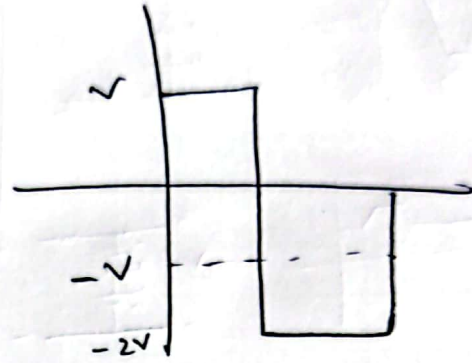
From $\frac{T}{2}$ to T , the direction is anticlockwise and diode is reverse biased so, it is open circuit.



Kirchhoff's law,

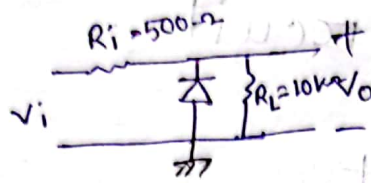
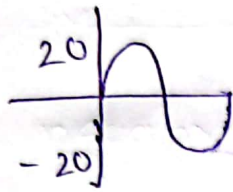
$$+v_i + v_o + V = 0$$

$$\begin{aligned} v_o &= -v_i - V \\ &= -V - V \\ &= -2V \end{aligned}$$



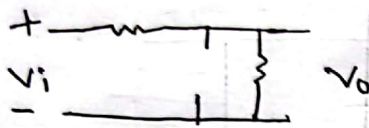
* Note: i) rms remains same
ii) peak value changes.

* Sketch the output wave form for the following network the diode is made of silicon.



Ans:

For Positive half cycle



20V will be divided into R_i and R_L (Drop in R_i and R_L)

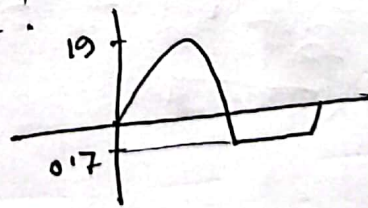
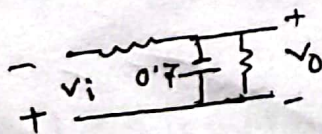
Given

$$v_i = 20V$$

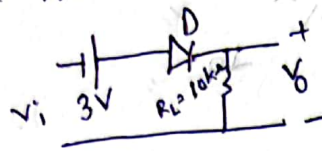
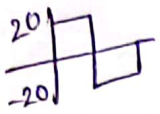
$$v_o = \frac{R_L}{R_i + R_L} \times v_i$$

$$= \frac{10}{10 + 0.5} \times 20 = 19.05V$$

For negative half cycle?

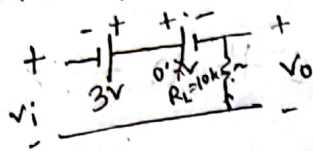


* Sketch the output waveform of the following circuit with given input waveform. The diode is made of silicon.



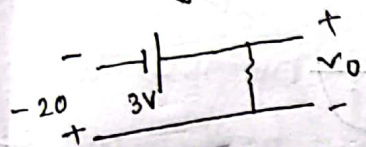
Solution:

For positive half cycle:



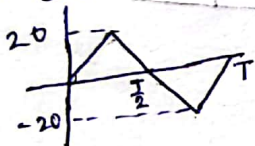
using KVL, $-3 + 0 + V_o - v_i = 0$
 $\Rightarrow V_o = 20 + 3 - 0 = 23V$

For negative half cycle:



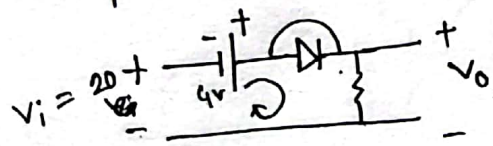
Waveform:

* Analyze the clipper network with given input and determine the output waveform of the given network.



Solution:

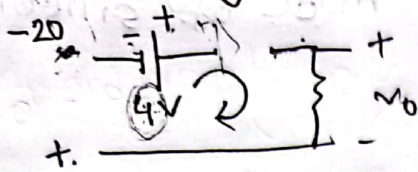
Here the output is directly against the resistor. In positive half cycle:



Using KVL,
 $-4 + V_o - v_i = 0$
 $V_o = 20 + 4 = 24V$

Here v_i and dc supply are both applying pressure to turn the diode on. The result is that the diode is 'on' state and short circuit.

For negative half cycle:



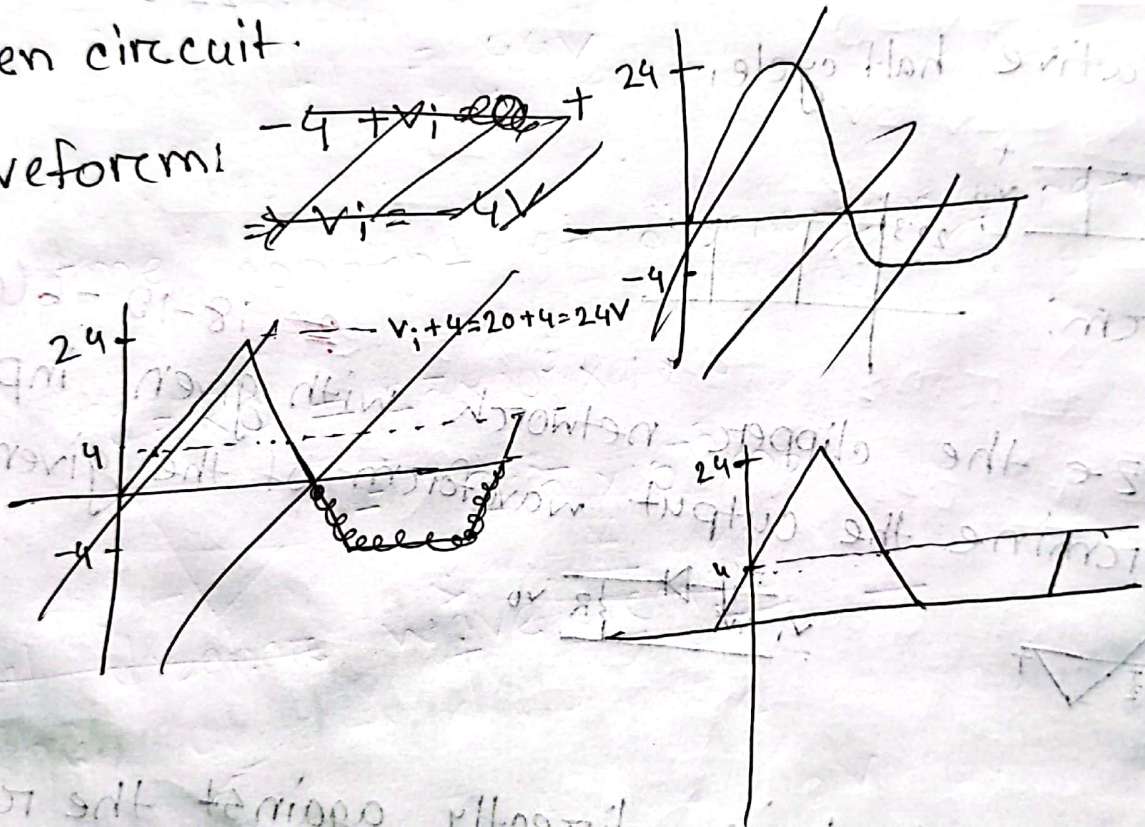
$$-4 + v_o + v_i = 0$$

$$-4 + v_o = -v_i$$

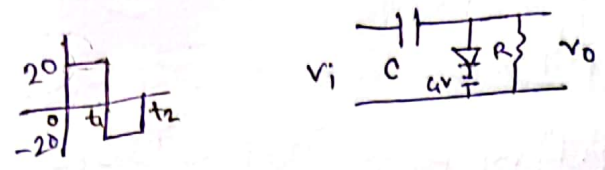
$$v_o = -v_i - 4$$

In negative supply, it would have to exceed the dc supply voltage of 4V before it could turn the diode off. So, voltage less than -4V is open circuit.

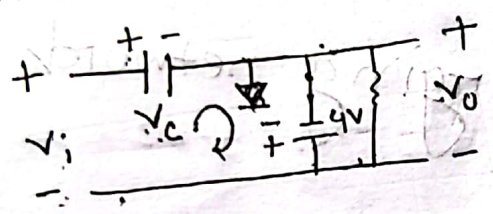
Waveform:



Analyze output voltage for given clamping circuit where $f = 2500 \text{ Hz}$, $R = 25 \text{ k}\Omega$



Solution:
The analysis will begin with period $0 \rightarrow t_1$ of the input signal since the diode is in its short circuit state.



$$V_o = V_i - V_C$$

$$= 20 - 16$$

$$= 4$$

$$V_C + V_o = V_i \Rightarrow 16 + 4 = 20$$

$$V_o = V_i - V_C$$

Applying KVL,

$$V_C - 4 + (-V_i) = 0$$

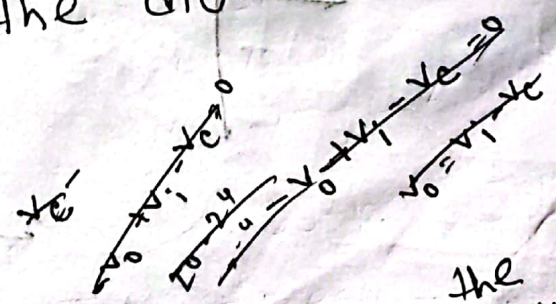
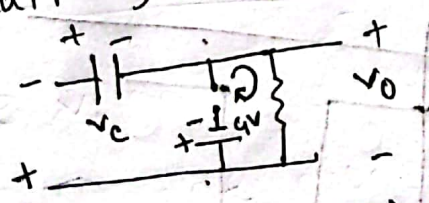
$$\Rightarrow V_C = V_i + 4$$

$$= 20 + 4$$

$$= 24 \text{ V}$$

The capacitor will therefore charge up to 24V.

For period $t_1 \rightarrow t_2$ the diode is in its open circuit state,



The open circuit is equivalent to the diode removed

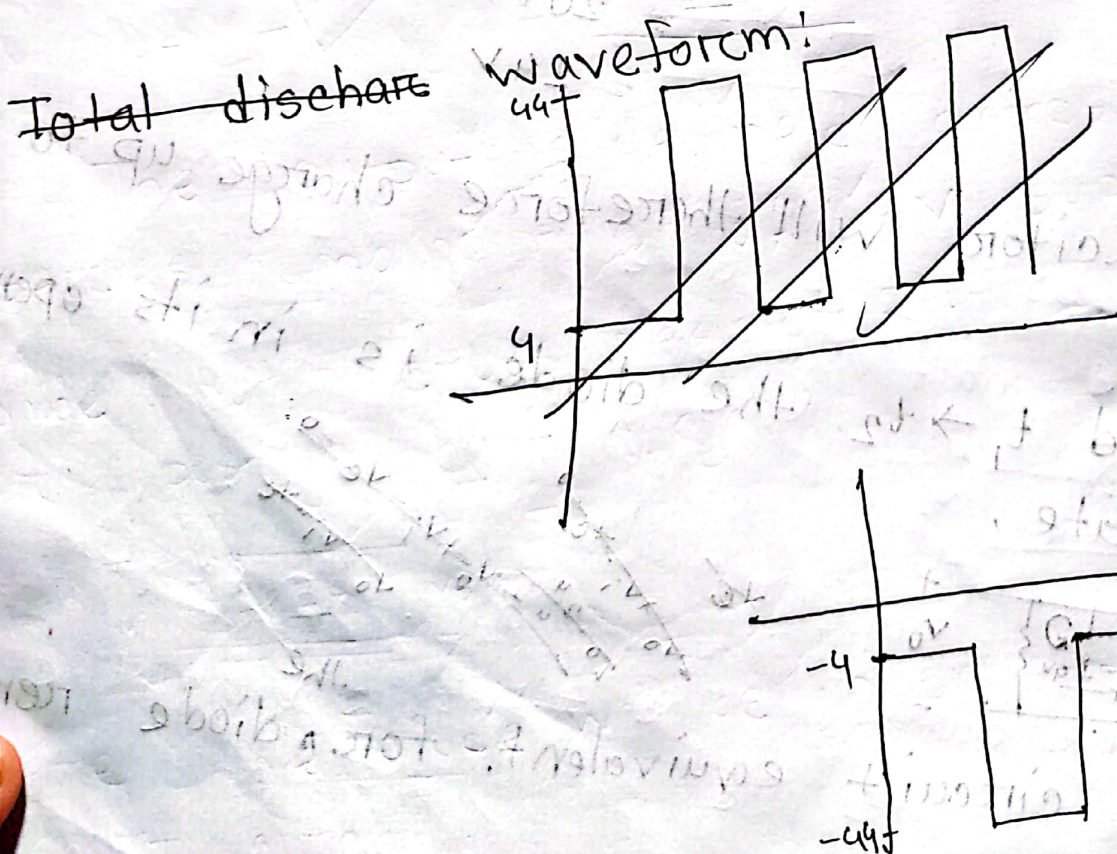
4V battery from having any effect on V_o and applying ~~K~~ KVL,

$$+V_c + V_o + V_i = 0$$

$$\Rightarrow V_o = -(V_i + V_c) \\ = -(20 + 24) \\ = -44V$$

The time constant of discharging network

$$\tau = RC = 25 \times 10^3 \times 1 \times 10^{-9} \\ = 2.5 \times 10^{-5} \text{ ms}$$



c) How resistors and capacitors are considered for clippers?

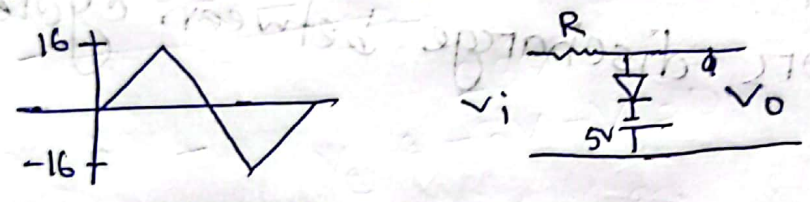
Ans: Resistors and capacitors are used to shift the dc level of a signal without altering its shape - capacitor enables the shifting of dc level in clamper ~~etc~~ circuit. It stores charge during operation of the circuit, which results in shifting signal's baseline up or down. Resistors are used to control the charging and discharging rate of the capacitor. It determines how quickly the capacitor can charge and how much the capacitor discharge between cycle.

2017-18-2
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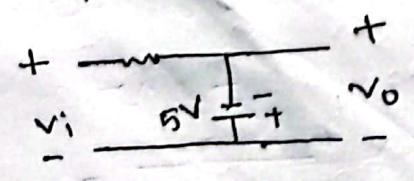
* 2a) Why clipper is used in electronic circuits?

Ans: Clippers are used in electronic circuit to limit or clip the amplitude of a signal to a specific level. ~~It~~ Clippers are used to modify the shape of an input waveform. Clipper can be combined with other component to shift DC level of a signal within a desired range.

* b) Determine the output waveform step by step of the given network.



Ans: The output is defined across the series combination of 5V supply and diode, not across resistor R.
For positive half cycle:



Here the diode will be in on state and short circuit.

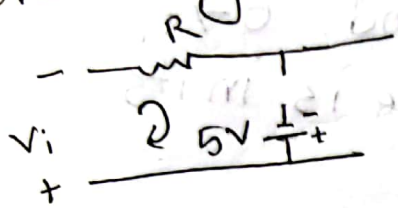
In this case, KVL,

$$v_o - 5V - v_i = 0$$

$$\Rightarrow v_o = v_i + 5$$

So, at the beginning the output voltage will be directly be 5V across 5V supply.

For negative half cycle,



In negative portion diode will be 'on' state for a good portion. Diode will on untill input voltage is less than 5V.

$$v_i \leq 5V$$

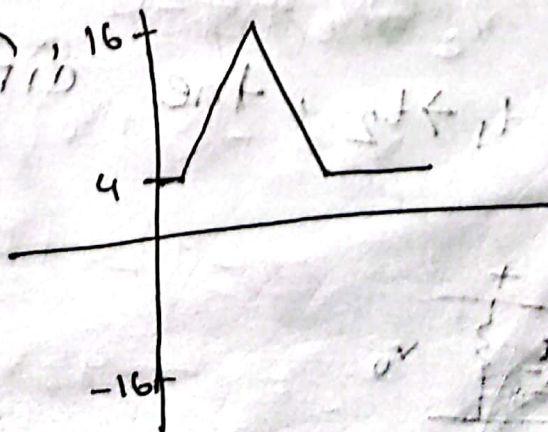
And for $v_i \geq 5V$, the diode will remain 5V.

Applying KVL,

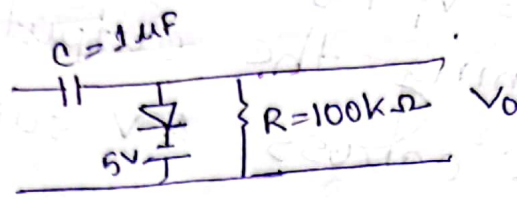
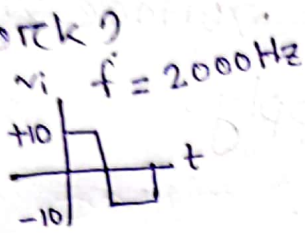
$$v_o - 5V + v_i = 0$$

$$v_o = -v_i + 5$$

So, the waveform,

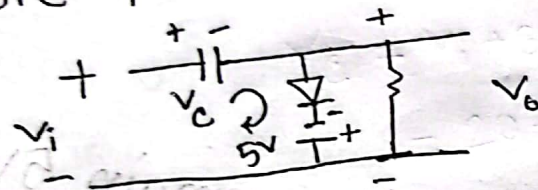


c) Analyze the output voltage and time constant in is analyzed in given clamping network?



Ans:

The analysis will begin with period $0 \rightarrow t_1$ of the input signal since the diode is in its short circuit state.
For positive half cycle:



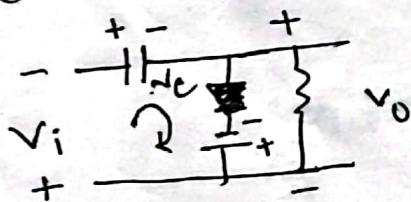
Using KVL, $v_c + (-5V) - v_i = 0$

$$v_c = v_i + 5V$$

$$= 10 + 5 = 15V$$

The capacitor will therefore charge up to 15V.

For period, $t_1 \rightarrow t_2$, the circuit will be open circuit.



The open circuit for the diode remove
~~4V~~ 5V battery from having any effect
 on v_o . applying KVL,

$$v_c + v_o + v_i = 0$$

$$\Rightarrow v_o = -v_i - v_c$$

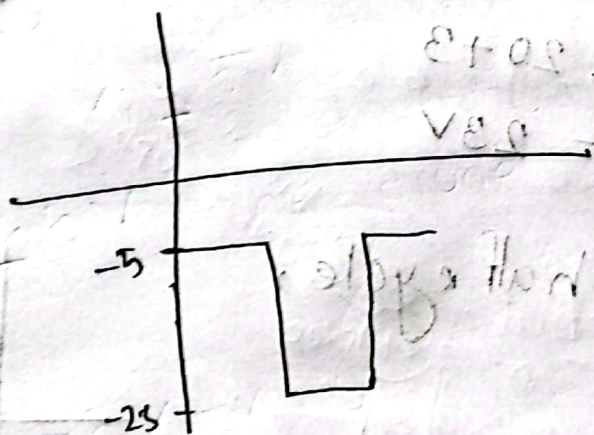
$$= -(v_i + v_c)$$

$$= -(v_i + 15)$$

$$= -(10 + 15)$$

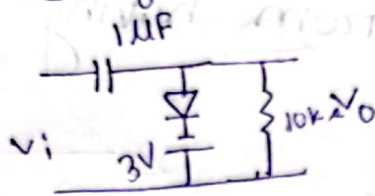
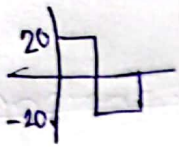
$$= -25V$$

∴ waveform:



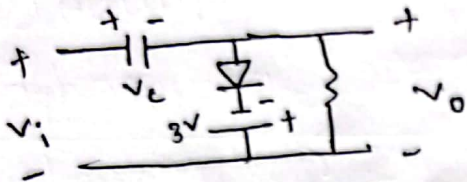
Time constant $\tau = RC = 100 \times 10^3 \times 1 \times 10^{-6} = 0.1 \text{ ms}$

3.a) Determine v_o for the given network 16-17-3



Ans:

For positive half cycle:



Using KVL from input side, for v_o ,

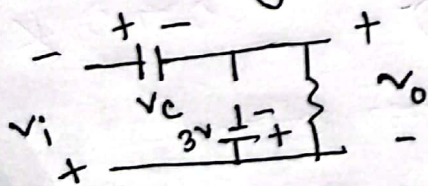
$$v_c - 3V - v_i = 0$$

$$\begin{aligned} v_c &= v_i + 3V \\ &= 20 + 3 \\ &= 23V \end{aligned}$$

$$v_c + v_o - v_i = 0$$

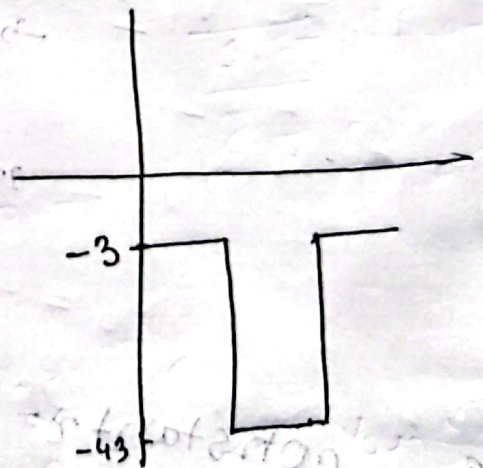
$$\begin{aligned} v_o &= v_i - v_c \\ &= 20 - 23 \\ &= -3V \end{aligned}$$

For negative half cycle,



Using KVL, $v_c + v_o + v_i = 0$

$$\begin{aligned} \Rightarrow v_o &= -(v_i + v_c) \\ &= -(20 + 23) \\ &= -43V \end{aligned}$$



- b) ~~Determine V_o for the given network~~
- c) Distinguish series and parallel clipper.

Series clipper	Parallel clipper
1. The diode is connected in series.	1. The diode is connected in parallel.
2. It is simple.	2. It is complex.
3. Power dissipation lower.	3. Power dissipation higher.
4. It removes the entire part of signal in reverse-biased.	4. It does not remove entire part of signal in reverse-biased.

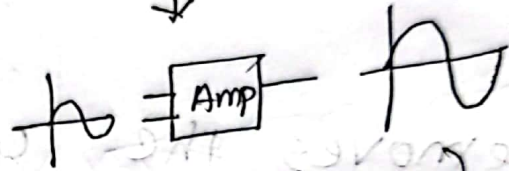
OP-Amp (Operational Amplifier)

Chapter - 2

Operational am
Operational Amplifier

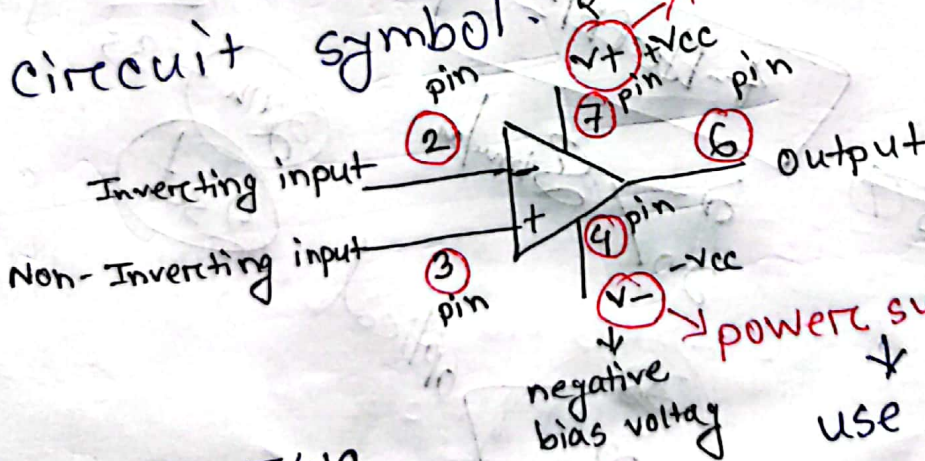
It is an integrated circuit that amplifies the difference in voltage between two inputs.

It is a early high-gain vacuum-tube amplifier designed to perform mathematical operations of addition, subtraction, multiplication, division, integration, differentiation. It is also used to amplifying signal.



Op-Amp এর total 8 terminal, 5 টা পিন
3 টা পিন।

* Circuit symbol:



[1, 5] → offset

[8] → not connected

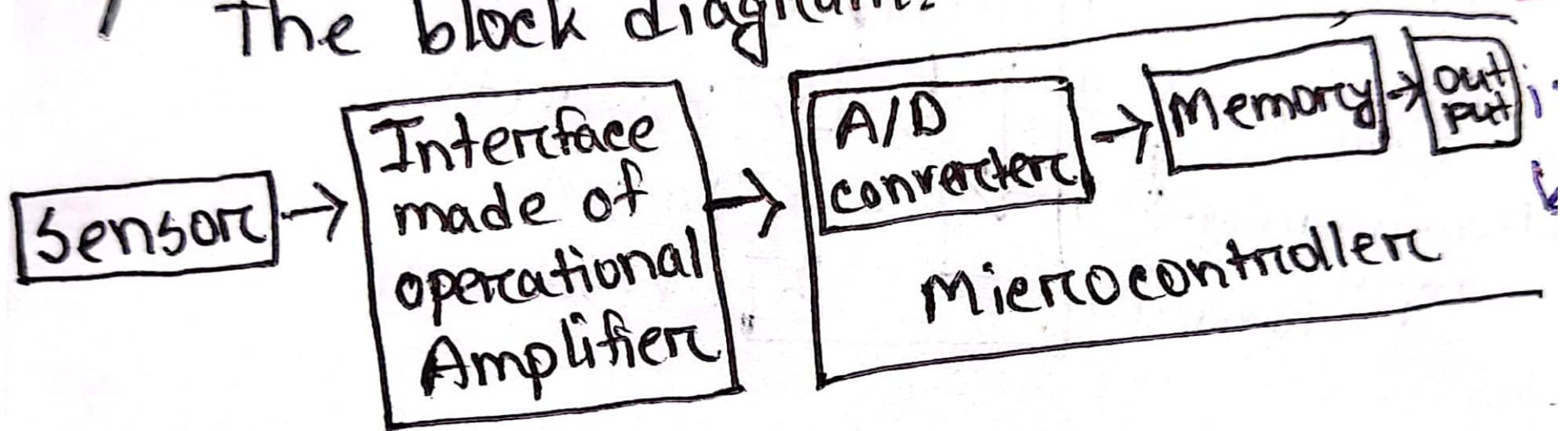
Fig: 741C

use করা হয় V_{out} এর value range এর সীমিত রাখার জন্য, Gain এর value অনেক বেছি থাকে তাই input voltage এর সাথে gain মিলে অনেক বড় output আসে, মোট reduce করতে

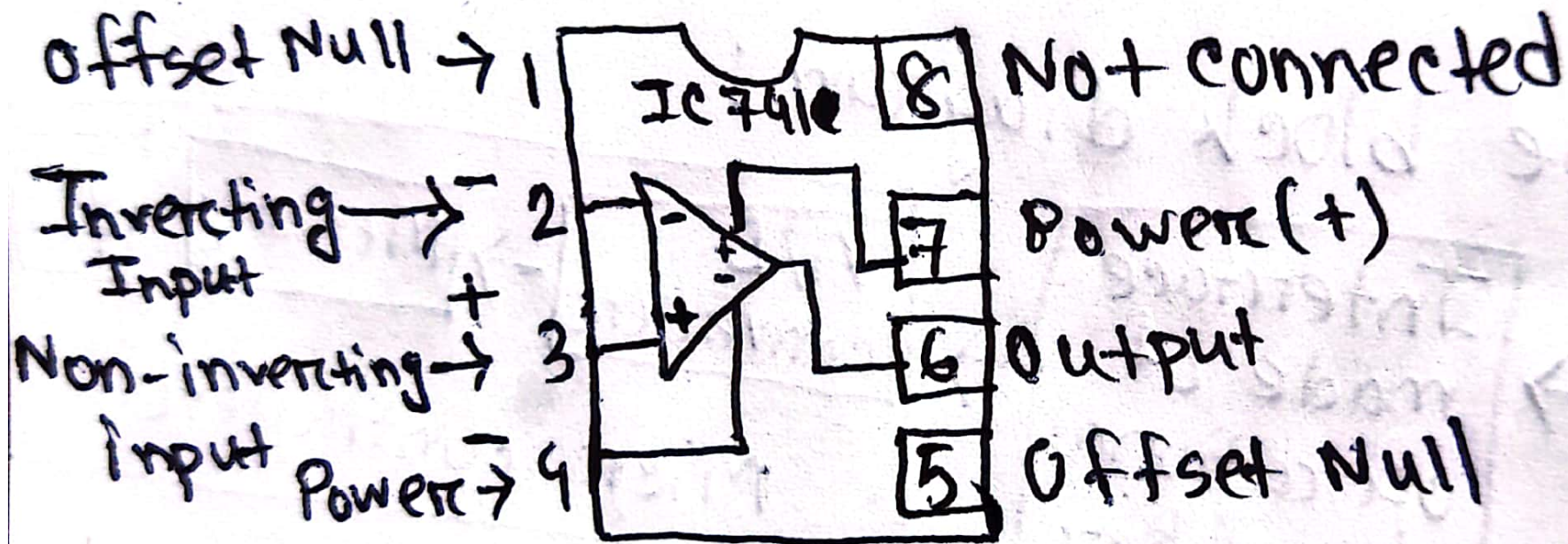
* Show the block diagram for using op-amp in sensor based acquisition.

Soln^o

The block diagram:

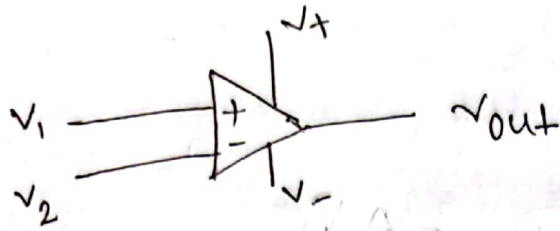


*Show the pin configuration of a typical op-amp - 2017-18-3(b)



Fig! Typical op-amp (741C)

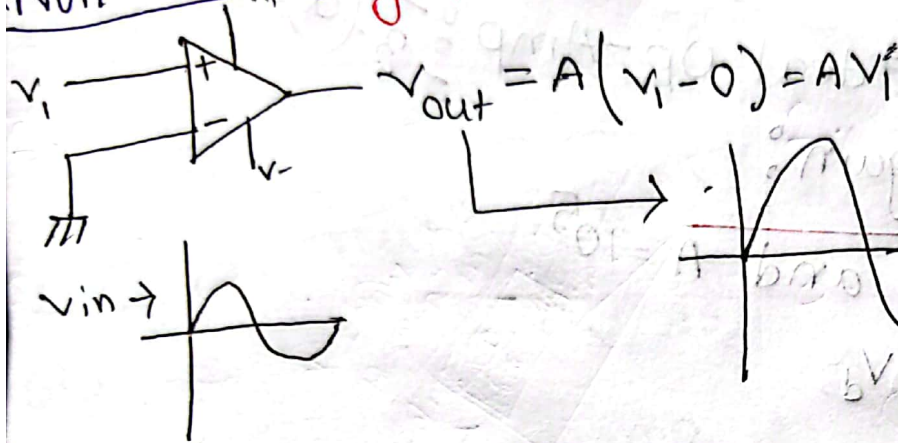
Differential amplifier



$$V_{out} = A(V_1 - V_2)$$

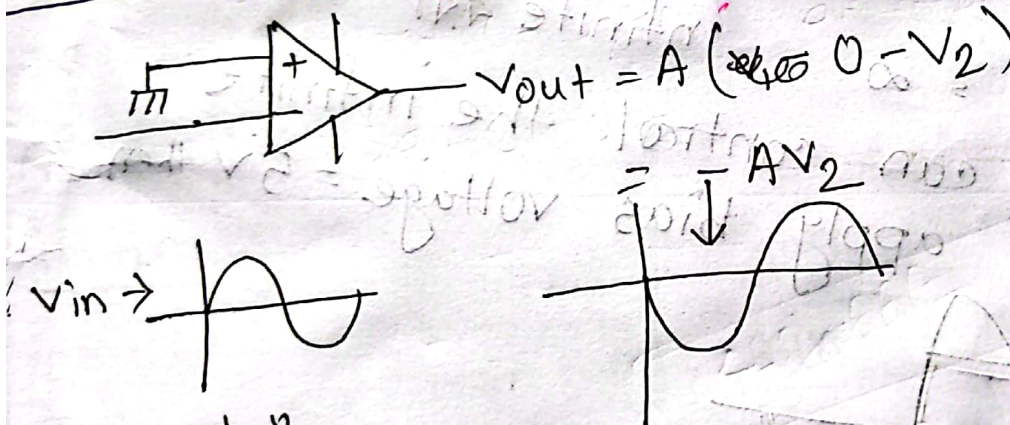
gain = $10^5 - 10^6$

Non-inverting



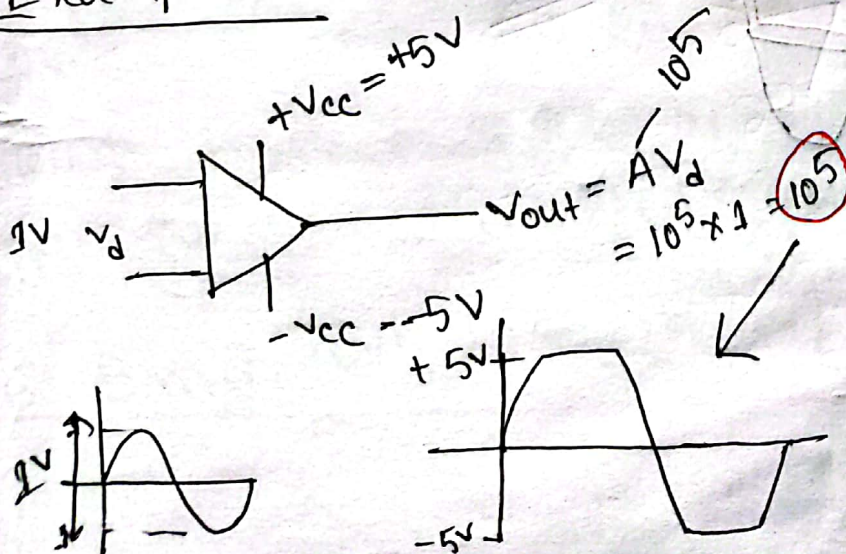
যেহেতু এখানে কোন
change/হচ্ছে না.
inverting
তাই non-inverting

* Inverting



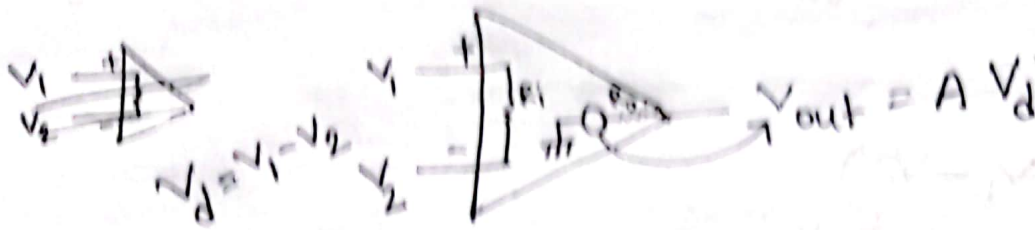
signal 180° invert
হচ্ছে তাই inverting

Example



value অনেক বেশি আছে তাই
bias voltage, +5V, -5V use করবে
এর জন্য wave 5V এ গিয়ে
ভাব বাক্য নেই.

* OP-Amp equivalent circuit:



* Characteristic of Ideal Op-Amp: 2017-18

1. Infinite Voltage gain:

Suppose, we $V_d = 1V$ and $A = 10^5$

We know, $V_{out} = A V_d$

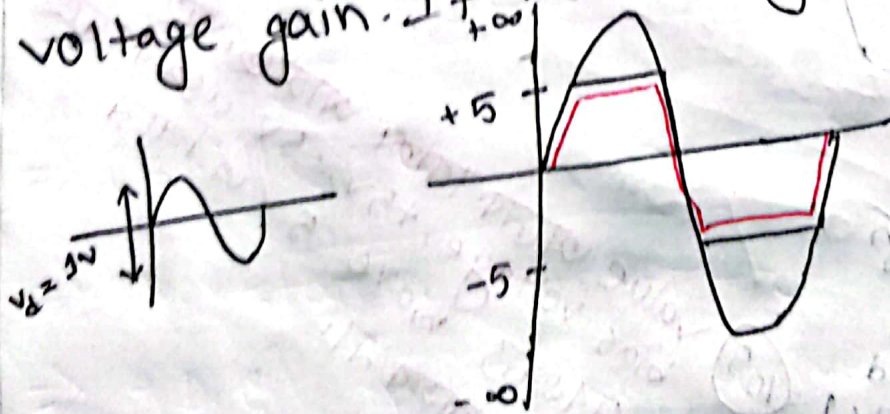
$= 10^5 \times 1$

$= 10^5 V$

$= \infty$

অনেক বেশি voltage তাই
infinite বরা

Using Bias voltage we can control the infinite voltage gain. If we apply bias voltage = 5V then



2. Infinite input impedance:
 $R_i = \infty$, so that current can't pass through V_1 and V_2 . Current will go direct to the circuit.

3. Output Impedance:
 $R_o = 0$

4. Slew rate is infinity:

slew rate $= \infty$ (V/ μ s)

How fast the op-amp is able to reach its final value.
কত ভায়েকটি output input কে track করতে পারবে।

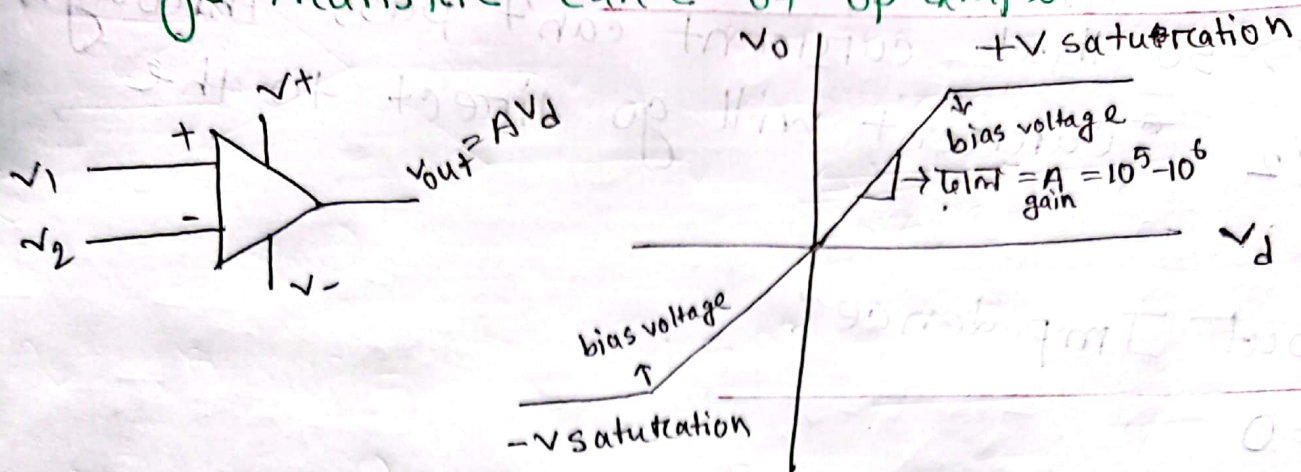
5. CMRR (Common Mode Rejection Ratio) is infinity.

$$CMRR = \frac{\text{Differential gain}}{\text{common mode gain}}$$

একটা op-amp কত ভালোভাবে common mode gain কে reject করে বা বাদ দিচ্ছে and differential gain কে কত ভালোভাবে gain করেছে।

6. Bandwidth is infinity.

* Voltage transfer curve of op-amp



* Application :

⇒ Several operation :

i) Addition

ii) Subtraction

iii) Integration

iv) Differentiation.

we can use them as:

⇒ Active filters

⇒ Oscillator

⇒ Waveform converter

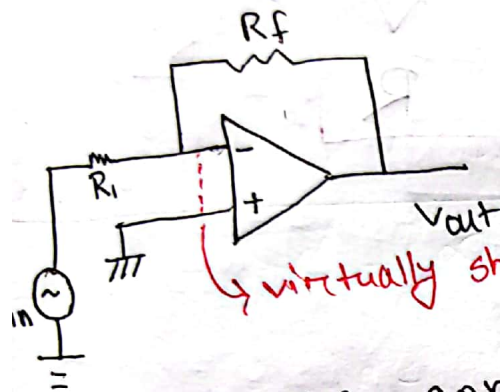
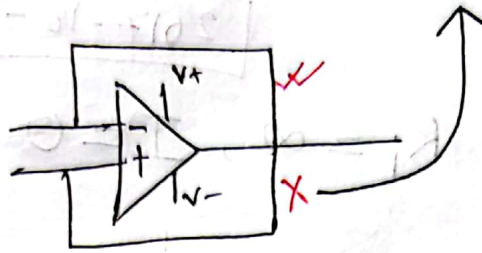
⇒ ADC and DAC converter

⇒ Voltage follower

⇒ voltage comparator

* Inverting op-amp and virtual ground concept:

* Positive voltage ∇ বা feedback use করলে
 unstable situation create হবে।



virtually shorted means both terminal will have some voltage.

As we connected signal to the inverting input (negative input) and ~~gain~~ positive input is connected to the ground, it is inverting op-amp.

Suppose, $V_{out} = 5V$, $A = 10^5$,

We know, $V_{out} = A V_d$

$$\Rightarrow V_d = \frac{V_{out}}{A}$$

$$= \frac{5}{10^5}$$

$$\frac{R_f}{R_1} = \frac{V_{out}}{V_d}$$

$$\frac{R_f}{R_1} = A$$

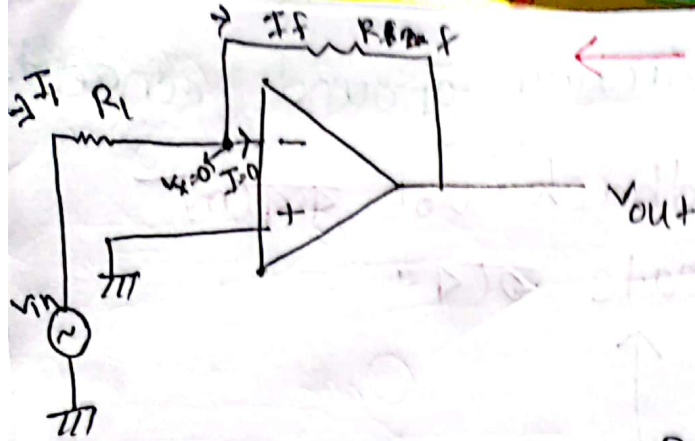
$$V_d = 0.00005V$$

$$V_d \approx 0$$

$$V^+ - V^- = 0$$

$$V^+ = V^-$$

it is called virtual ground.



← * How voltage gain is obtained for inverting amplifier?

~~2018-19~~
2017-18 - 4 (c)

As, in op-amp, $R_i = \infty$, $I_i = 0$

$$V^+ = V^- = 0$$

$$* I = \frac{V}{R}$$

$$I_i = I_f$$

$$\Rightarrow \frac{V_{in} - V_x}{R_i} = \frac{V_x - V_{out}}{R_{f}}$$

$$\Rightarrow \frac{V_{in}}{R_i} = \frac{-V_{out}}{R_{f}}$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = - \frac{R_f}{R_i}$$

$$\Rightarrow A = - \frac{R_f}{R_i}$$

এটা indicate করছে signal input এ যা থাকবে output এ 180° change হয়ে যাবে।

Suppose, $R_f = 200k\Omega$

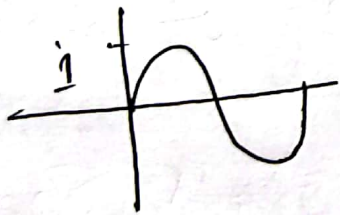
$R_1 = 20k\Omega$

$$\therefore A = -\frac{R_f}{R_1} = -\frac{200}{20} = -10$$

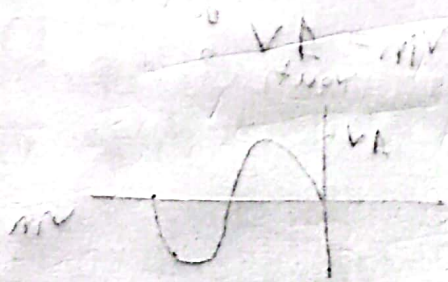
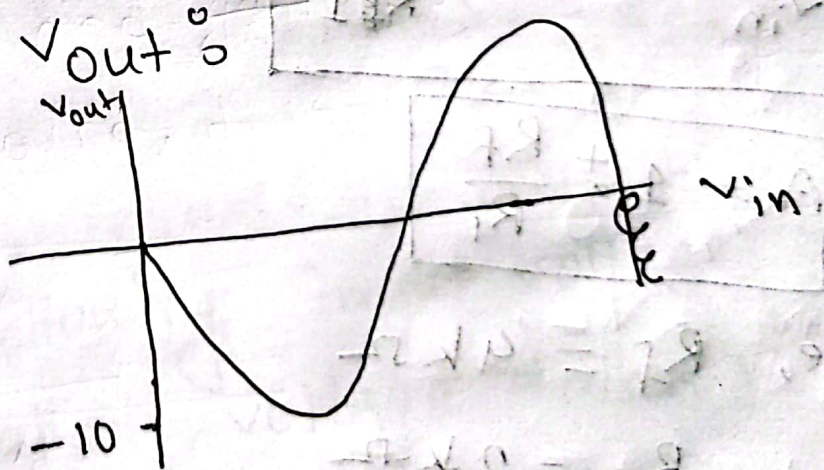
$$\frac{V_{out}}{V_{in}} = 10$$

$$\Rightarrow V_{out} = -10 \times V_{in}$$

If $V_{in} = 1V$:

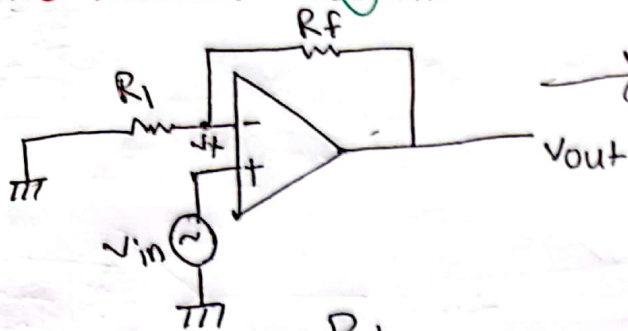


V_{out}

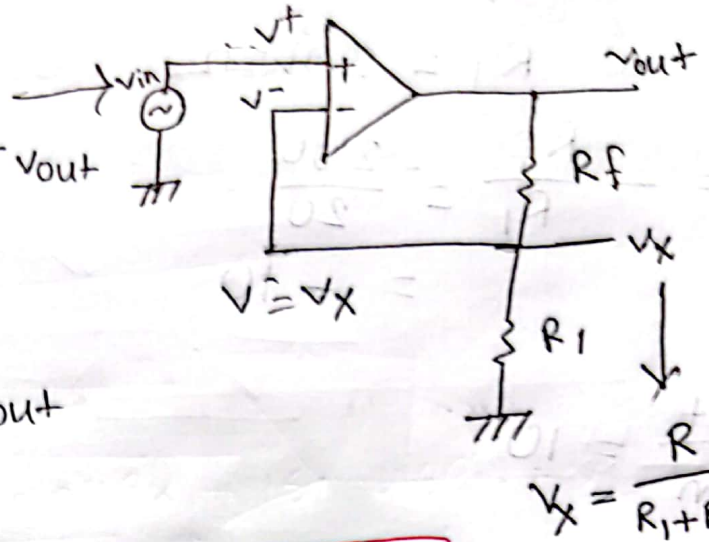


*Non-Inverting op-amp & negative feedback use

2017-18
-3(c)



$$V^+ = V^- = V_x = \frac{R_1}{R_1 + R_f} \times V_{out}$$



$$V_x = \frac{R}{R_1 + R_f} \times V_{out}$$

$$\Rightarrow V_{in} = \frac{R_1}{R_1 + R_f} \times V_{out}$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_1}$$

$$\Rightarrow A = 1 + \frac{R_f}{R_1}$$

Suppose, $R_f = 4k\Omega$
 $R_1 = 2k\Omega$

$$A = 1 + \frac{4}{2} = 1 + 2 = 3$$

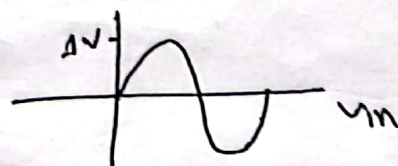
$$\frac{V_{out}}{V_{in}} = 3$$

$$\Rightarrow V_{out} = 3 \times V_{in} = 3 \times 1 = 3V$$

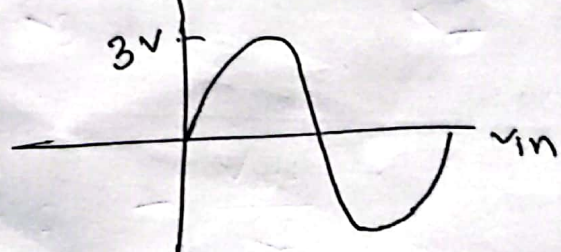
2018-19-3(c)

*Draw the circuit for a non-inverting amplifier and explain its operation

if $V_{in} = 1V$



then $V_{out} = 3V$



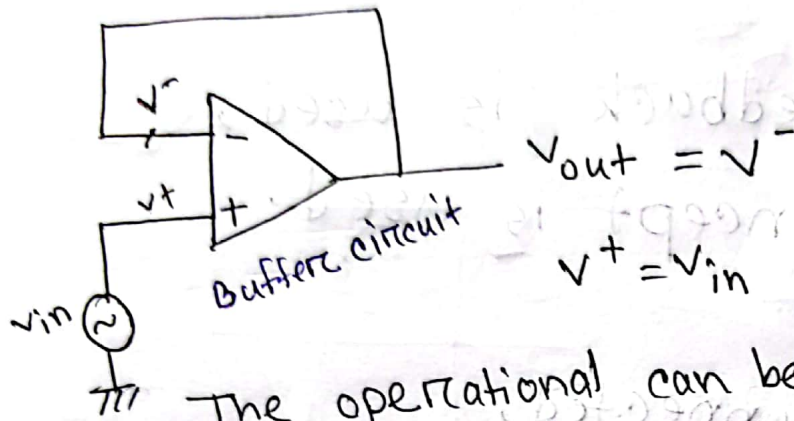
* Op-amp as a voltage follower / op-amp as a buffer %

* Draw the circuit of a Op-Amp as voltage follower and find an expression for its voltage gain.

2018-19-3(b)

$$R_f = 0 \Omega$$

$$R_i = \infty$$



The operational can be used to design a voltage follower if we make $R_f = 0$ and $R_{i1} = \infty$. Because of feedback, we have virtual short between inverting and non-inverting terminals, we can say that

$$\Rightarrow v_{in} = v_{out}$$

So, input voltage is equal to output voltage. Output voltage is following input voltage. So, that's why it is called voltage follower.

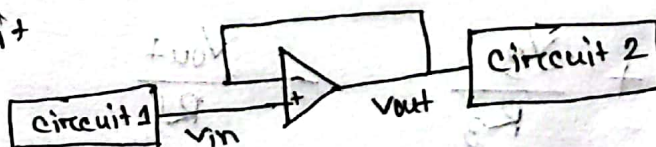
* Main characteristic of buffer circuit:

i) very high input impedance, $Z_{in} = \infty$

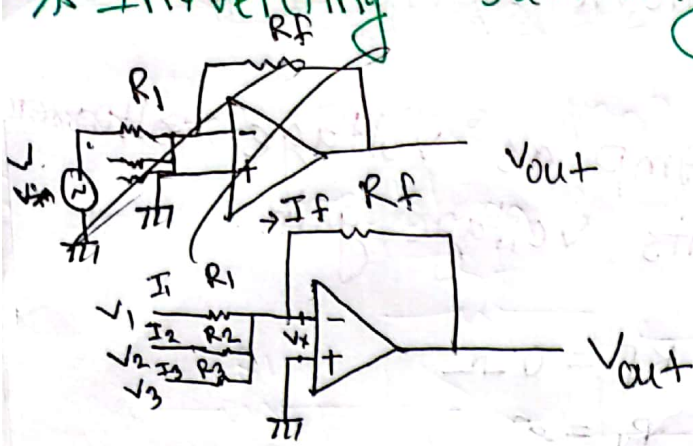
ii) $v_{out} = v_{in}$

Use of buffer circuit:

Using buffer circuit we can isolate two different circuit at the same time. we can ensure that whatever voltage that is appearing at the output of one circuit will appear at input of other circuit



* Inverting Summing Amplifier



When negative feedback is used, virtual ground concept is used.

$$V^+ = V^-$$

As, V^+ is ground connected,

$$V^+ = 0 = V^-$$

$$V_x = 0$$

Applying KVL,

$$I_1 + I_2 + I_3 = I_f$$

They are entering in point X

exiting from point X

$$\Rightarrow \frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_{out}}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_{out}}{R_f}$$

$$V_{out} = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$

If $R_1 = R_2 = R_3$,

$$V_{out} = - \frac{R_f}{R} (V_1 + V_2 + V_3)$$

$$= - (V_1 + V_2 + V_3) \quad [\text{If } R_f = R]$$

↓
As it is inverting amplifier it will change wave 180° in output.

* Scaling operations

$$V_{out} = - (A V_1 + B V_2 + C V_3)$$

$\downarrow$ $\downarrow$ $\downarrow$
 $\frac{R_f}{R_1}$ $\frac{R_f}{R_2}$ $\frac{R_f}{R_3}$

* Averaging operations

$$R_1 = R_2 = R_3 = R$$

$$V_{out} = - \frac{R_f}{R} (V_1 + V_2 + V_3)$$

$$\text{If, } \frac{R_f}{R} = \frac{1}{n} = \frac{1}{3}$$

$$= - \left(\frac{V_1 + V_2 + V_3}{3} \right)$$

* Use of inverting adders

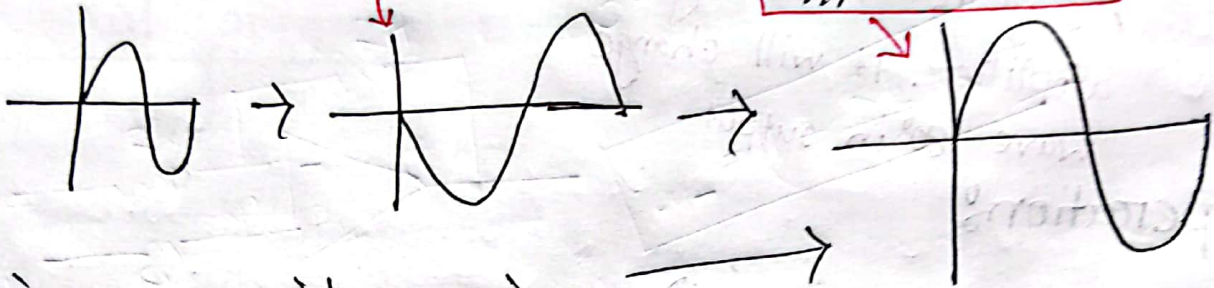
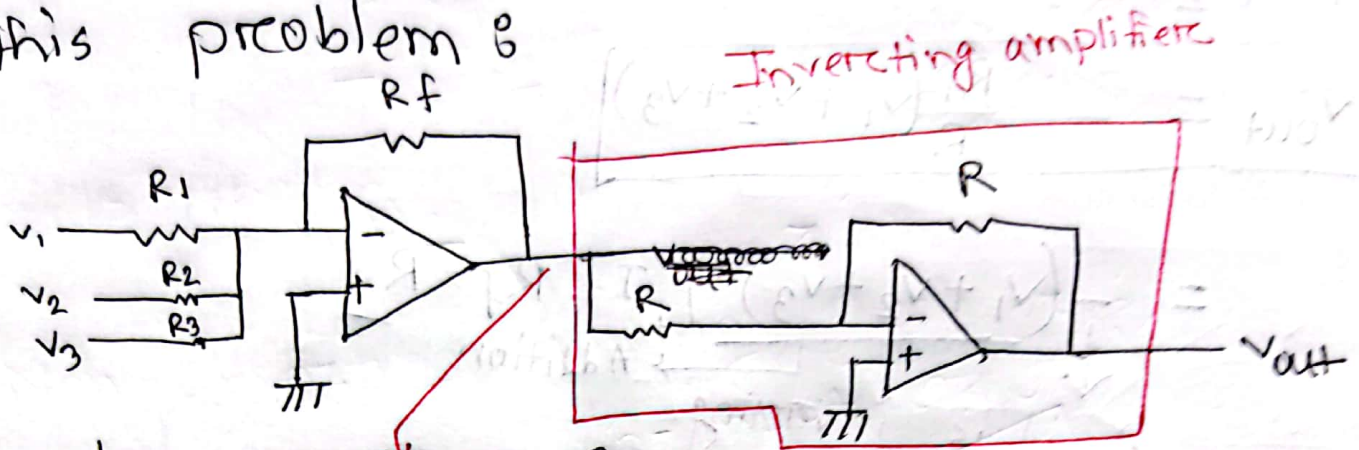
⇒ Summing, averaging, scaling

⇒ providing DC offset.

⇒ DAC (Digital to analog converter)

⇒ Audio mixer

* ~~As~~ ~~inverting amplifier~~ As inverting amplifier invert wave 180° , to ~~reduce~~ remove this problem &

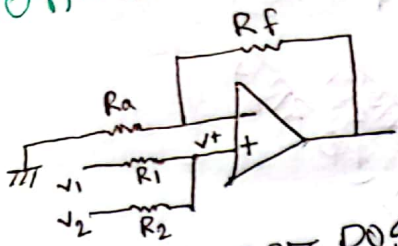


Using unity gain,

$$A = \frac{R}{R} = 1$$

So, if we give a magnitude sine wave we will get amplified sine wave in the same phase without inverting it.

*Non-Inverting Summing Amplifier (2-input এর জন্য)



$$V_{out} = \left(1 + \frac{R_f}{R_a}\right) V^+$$

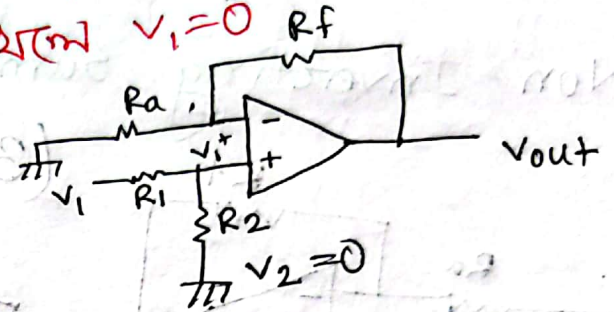
Applying superposition

→ at a time একটি source কাজ করা, v_1 কাজ করলে $v_2 = 0$, v_2 active রাখলে $v_1 = 0$

When, $v_2 = 0$

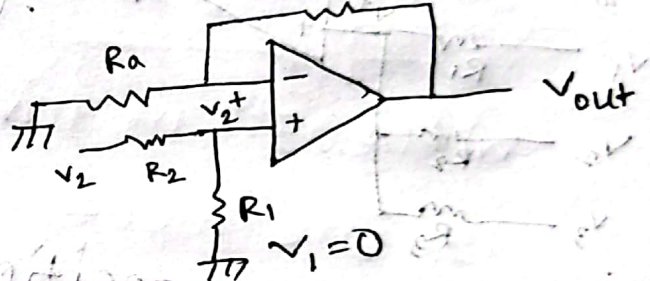
$$V_1^+ = \frac{R_2}{R_1 + R_2} \times V_1$$

voltage divider rule



When, $v_1 = 0$

$$V_2^+ = \frac{R_1}{R_1 + R_2} \times V_2$$



$$V^+ = V_1^+ + V_2^+ = \frac{R_2}{R_1 + R_2} \times V_1 + \frac{R_1}{R_1 + R_2} \times V_2$$

when, both v_1 and v_2 works,

$$V_{out} = \left(1 + \frac{R_f}{R_a}\right) V^+$$

$$= \left(1 + \frac{R_f}{R_a}\right) \left(\frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2 \right)$$

$$V_{out} = \left(1 + \frac{R_f}{R_a}\right) \left(\frac{V_1 + V_2}{2} \right) \left[\text{If } R_1 = R_2 = R \right]$$

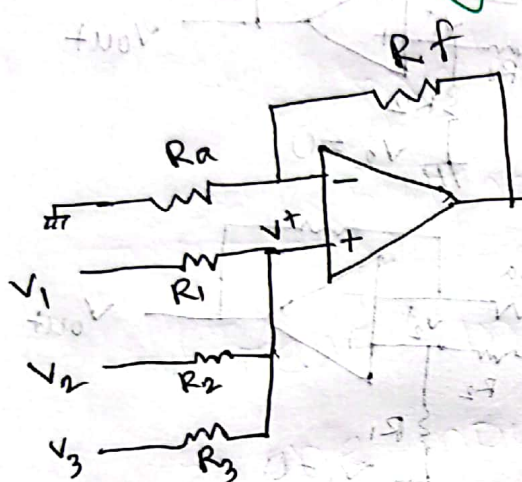
$$V_{out} = (1+1) \left(\frac{V_1 + V_2}{2} \right) [I f R_f = R_a]$$

$$= 2 \cdot \frac{V_1 + V_2}{2}$$

$$= V_1 + V_2$$

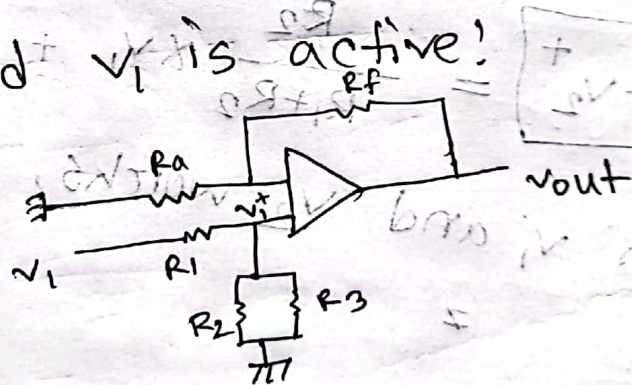
input complexity ৩০ ৩০

*Non-Inverting Summing Amplifier :
(3-input এর জন্য)



$$V_{out} = \left(1 + \frac{R_f}{R_a} \right) V_+$$

Applying superposition ;
when $V_2 = V_3 = 0$ and V_1 is active!



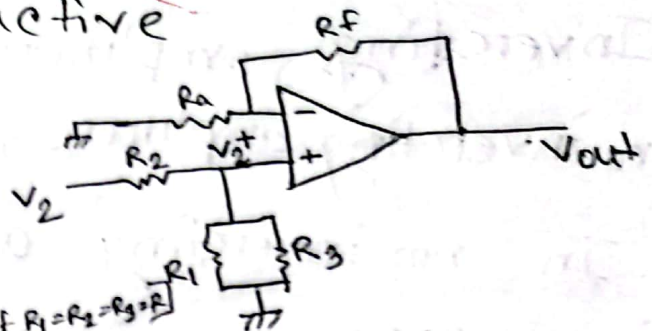
$$V_+ = \frac{R_2 || R_3}{R_1 + R_2 || R_3} \times V_1$$

$$= \frac{\frac{R_2 \times R_3}{R_2 + R_3}}{R_1 + \frac{R_2 \times R_3}{R_2 + R_3}} \times V_1$$

$$= \frac{\frac{R^2}{2R}}{R + \frac{R^2}{2R}} \times V_1 = \frac{\frac{R^2}{2R}}{\frac{2R^2 + R^2}{2R}} \times V_1 = \frac{R^2}{2R^2 + R^2} \times V_1 = \frac{R^2}{3R^2} \times V_1 = \frac{1}{3} V_1$$

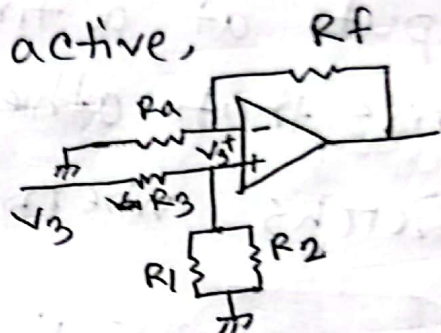
When, $V_1 = V_3 = 0$, V_2 is active

$$V_2^+ = \frac{R_1 \parallel R_3}{R_2 + R_1 \parallel R_3} \times V_2$$



$$= \frac{\frac{R_1 R_3}{R_1 + R_3}}{R_2 + \frac{R_1 R_3}{R_1 + R_3}} \times V_2 = \frac{V_2}{3} \quad [\text{If } R_1 = R_2 = R_3 = R]$$

When $V_1 = V_2 = 0$, V_3 is active,



$$V_3^+ = \frac{R_1 \parallel R_2}{R_3 + R_1 \parallel R_2} \times V_3$$

$$= \frac{\frac{R_1 R_2}{R_1 + R_2}}{R_3 + \frac{R_1 R_2}{R_1 + R_2}} \times V_3 = \frac{V_3}{3} \quad [\text{If } R_1 = R_2 = R_3 = R]$$

Suppose, $R_1 = R_2 = R_3 = R$

$$\therefore V_{out} = V_1^+ + V_2^+ + V_3^+ = \frac{V_1}{3} + \frac{V_2}{3} + \frac{V_3}{3}$$

$$= \frac{V_1 + V_2 + V_3}{3}$$

$$\therefore V_{out} = \left(1 + \frac{R_f}{R_a}\right) \left(\frac{V_1 + V_2 + V_3}{3}\right)$$

If $1 + \frac{R_f}{R_a} = 3$ then,

$$V_{out} = 3 \times \frac{V_1 + V_2 + V_3}{3}$$

$$= V_1 + V_2 + V_3$$

* Inverting amplifier is better than non-inverting amplifier, because:

1. In non-inverting amplifier, when there is more than one input, suppose three input then when it needs one input at a time to work but it seems that other resistance still works when other voltage is zero. whereas, in inverting amplifier there is one input and no other resistance of other ^{input} interferes.

* अब equation टाइम बूझा यावे।

* Difference between inverting and non-inverting amplifier?

Inverting Amplifier

1. A type of operational amplifier which generates an amplified output signal that is 180° out of phase with input signal is called inverting amplifier.

2. The input signal is applied at the negative terminal.

3. Phase difference is 180°

4. Input and output signals are out of phase.

5. Feedback is provided at the same terminal at which input is applied.

6. Positive terminal is connected to the ground.

7. Closed loop gain,

$$A = -\frac{R_2}{R_1}$$

Non-inverting Amplifier

1. A type of operational amplifier which generates an amplified output signal that is in phase with input signal is called non-inverting amplifier.

2. The input signal is applied at the positive terminal.

3. Phase difference is 0° .

4. Input and output signal are in phase.

5. Feedback and input signal are connected to different terminal.

6. Negative terminal is connected to the ground.

7. Closed loop gain,

$$A = 1 + \frac{R_2}{R_1}$$

Inverting

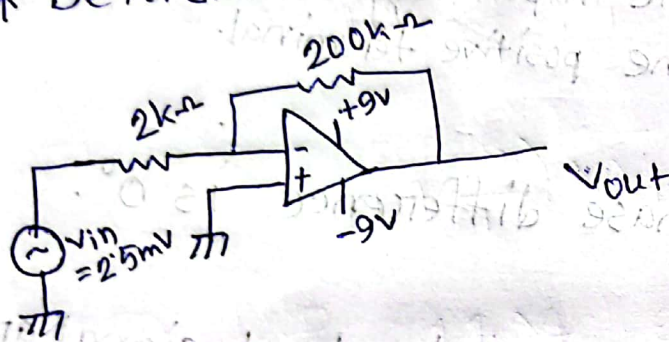
Application: in trans-resistance amplifiers, at the output stage in the circuits designed with different sensors, integrators, circuit, phase shifter.

Non-inverting

Application: in circuits that require high impedance, isolation of cascaded circuit, changing gain considerations, voltage followers.

* Inverting Op-Amp Math Problems

* Determine the output voltage:



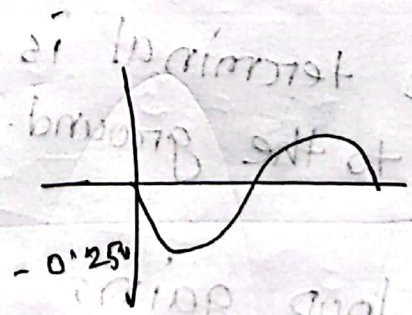
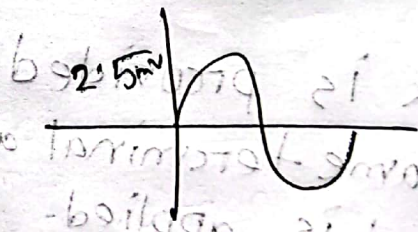
Soln:

We know, $V_{out} = A \times V_{in}$

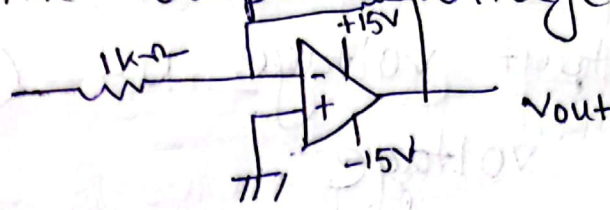
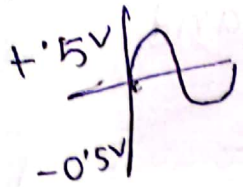
$$A = -\frac{R_f}{R_i}$$

$$\therefore A = -\frac{R_f}{R_i} = -\frac{200k}{2k} = -100$$

$$V_{out} = -100 \times 2.5mV = -0.25V$$



* Determine the output voltage;



Soln^o
We know, $V_{out} = A \times V_{in}$

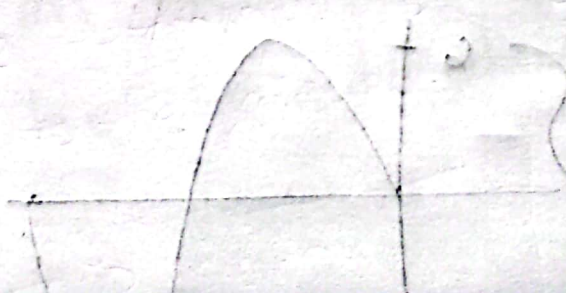
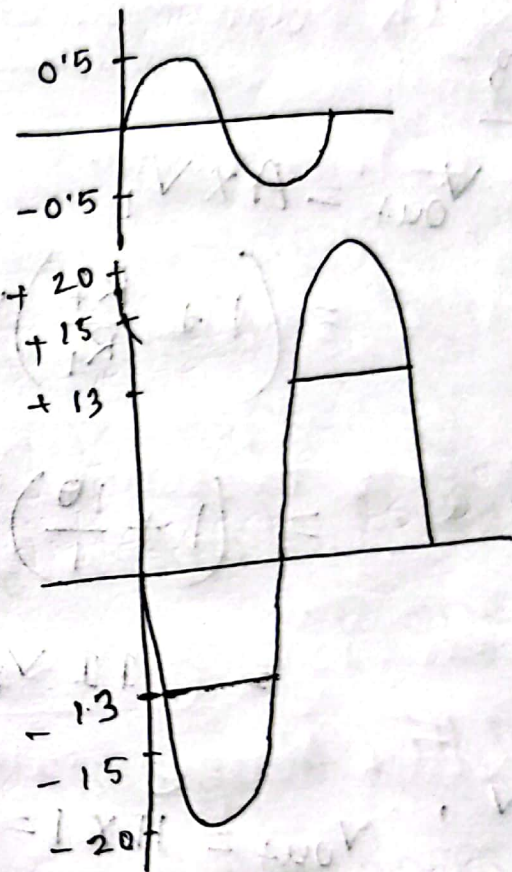
$$A = -\frac{R_f}{R_i}$$

$$= -\frac{40}{1}$$

$$= -40$$

$$V_{out} = A \times V_{in} = -40 \times 0.5$$

$$= -20V$$



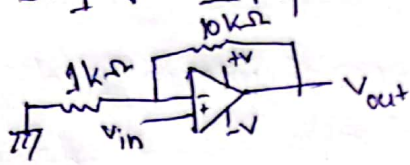
$$A = \frac{R_f}{R_i} = 1$$

$$(1 + \frac{R_f}{R_i}) = 2$$

* Non-Inverting Op-Amp Math problem

* Find out the output voltage of ① 1V and

② -1V Input voltage.



Soln 8

$$\textcircled{1} V_{out} = A \times V_i$$

$$= \left(1 + \frac{R_f}{R_i}\right) V_i$$

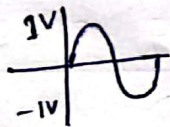
$$= \left(1 + \frac{10}{1}\right) \times V_i$$

$$= 11 V_i$$

$$V_i = 1V, \quad V_{out} = 11 \times 1 = 11V$$

$$\textcircled{2} V_{out} = 11 \times (-1) = -11V$$

* Find the peak to peak output voltage:



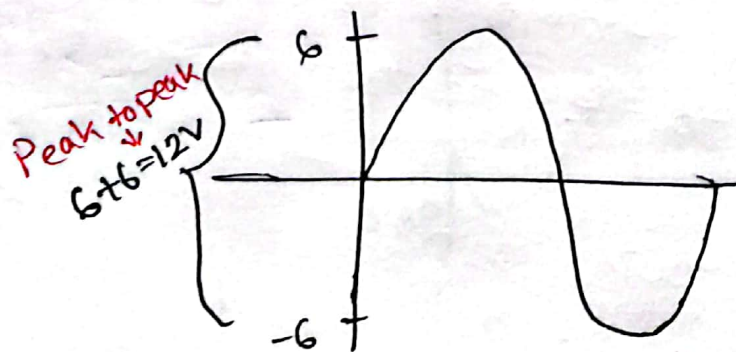
Soln:

$$A = 1 + \frac{R_f}{R_i}$$

$$= \left(1 + \frac{5}{1}\right)$$

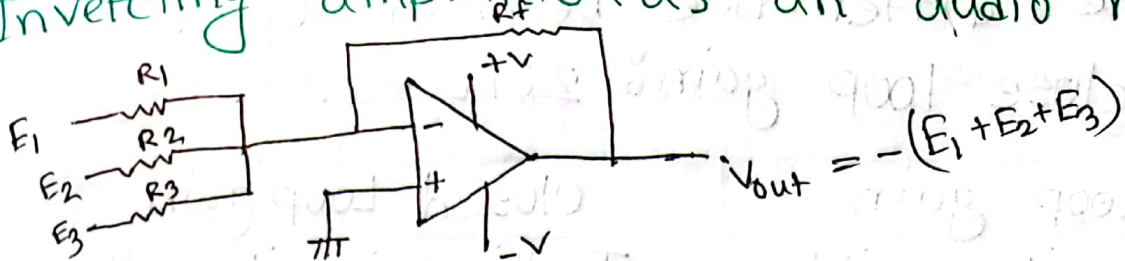
$$V_{out} = 6 \times V_i$$

$$= 6 \times 1 = 6V$$



Ans: 12V

* Inverting amplifier as an audio mixer



Here all input currents flow through feedback resistor R_f . This means I_1 does not affect I_2 or I_3 . More generally, the input currents do not affect one another because each ^{'sees'} ground potential at the summing node. Therefore, input voltages do not interact. This feature is especially desirable in audio mixers, for example, E_1, E_2, E_3 are replaced by microphones. Then if one microphone is carrying guitar music, it will not come out of a second microphone facing singer. If a resistor is ~~ad-~~ installed between microphone and input resistor, their relative volume can be adjusted and added. A weak singer can then be heard above a very loud guitar.

* Show the difference between open loop gain and close loop gain: 2018-19-4(a)

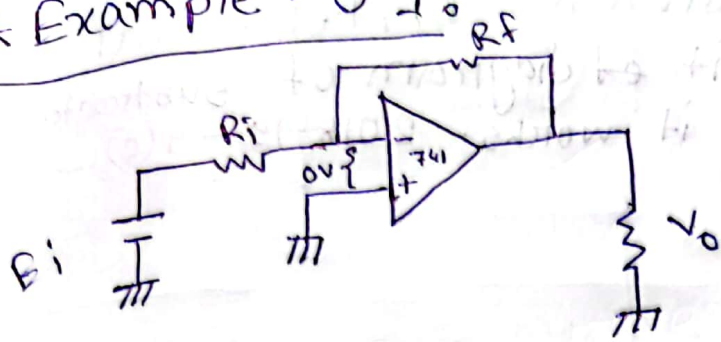
open loop gain

1. In this circuit possible feedback connection of output terminal to input terminal are left open.
2. It is simple
3. It gets response fast
4. Open loop gain value is very large
5. It is uncontrolled and unstable.
6. It is not used directly in practical application.

closed loop gain

1. In this circuit possible feedback connection of output terminal to input terminal are closed or connected.
2. It is complex
3. It gets response slow
4. Closed loop gain value is small
5. It is controlled and stable.
6. It is used in practical application.

* Example - 3.1



Let $R_f = 100k\Omega$, $R_i = 10k\Omega$ and $E_i = 1V$. Calculate a) I

b) V_o c) A_{CL}

a) We know, $I = \frac{E_i}{R_i} = \frac{1}{10 \times 10^3} = 0.1mA$

$$b) V_o = - \frac{R_f}{R_i} \times E_i = - \frac{100}{10} \times 1 = -10V$$

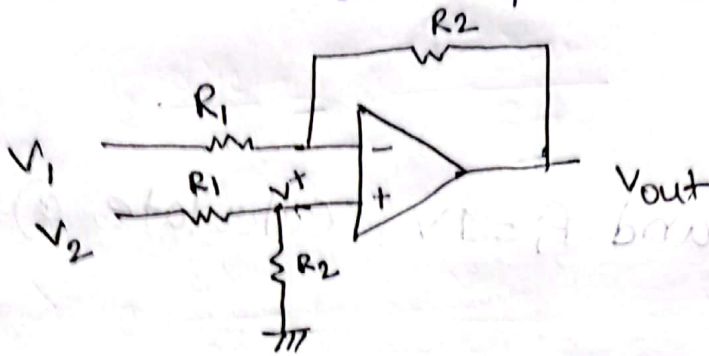
$$c) A_{CL} = - \frac{R_f}{R_i} = - \frac{100}{10} = -10$$

$$-1V \times \left(\frac{100}{10} + 1 \right) = -11V$$

$$-1V \times \left(\frac{100}{10} + 1 \right) = -11V$$

$$-1V \times \left(\frac{100}{10} + 1 \right) = -11V$$

* OP-Amp as a subtractor / Differential amplifier :
 * Draw the circuit diagram of subtractor and explain how it works. 2018-19-4(c)



This circuit is the combination of inverting and non-inverting configuration. Applying superposition theorem, for V_{out}

Let us consider, $V_2 = 0$, $V_1 = 0$
 using voltage divider rule,

$$V_2 = V^+ = \frac{R_2}{R_1 + R_2} \times V_2$$

$$V_{out} = \left(1 + \frac{R_2}{R_1}\right) V^+$$

$$= \left(1 + \frac{R_2}{R_1}\right) \times \left(\frac{R_2}{R_1 + R_2}\right) \times V_2$$

$$= \frac{(R_1 + R_2)}{R_1} \times \frac{R_2}{(R_1 + R_2)} \times V_2$$

$$= \left(\frac{R_2}{R_1}\right) V_2$$

$$V_2 = 0$$

$\nabla \neq$

$$V^- = -\frac{R_2}{R_1} V_1$$

$$V_{out} = V^+ + V^-$$

$$= \frac{R_2}{R_1} V_2 - \frac{R_2}{R_1} V_1$$

$$V_{out} = \frac{R_2}{R_1} (V_2 - V_1)$$

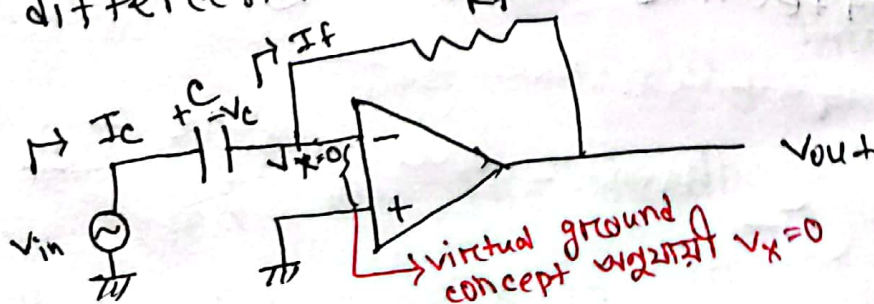
* Show circuit diagram of integrator and differentiator and explain how it works.

2018-19-4(b)

* OP-amp as differentiator

Input ও Output এ যা পার তা হবে input এর differentiation আকারে

If we replace input resistor of inverting configuration with a capacitor then we get a simple differentiator circuit.

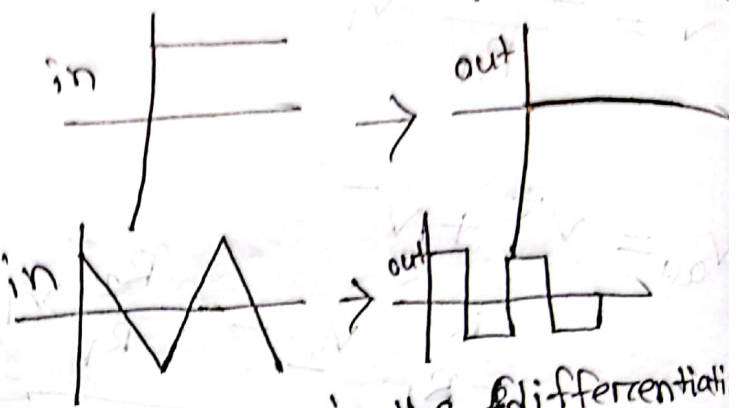
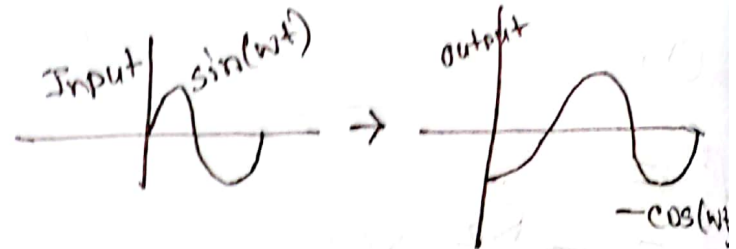


Applying KCL,
 $I_c = I_f = 0$
 $I_c = I_f$

$$C \frac{dV_c}{dt} = \frac{0 - V_{out}}{R_f}$$

$$\Rightarrow C \frac{dV_{in}}{dt} = - \frac{V_{out}}{R_f}$$

$$\Rightarrow \boxed{V_{out} = -R_f C \frac{dV_{in}}{dt}}$$



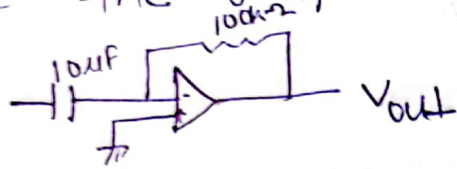
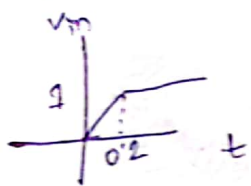
So, the output is the differential of input voltage times $R_f C$ factor.

The current flowing through the capacitor I_c and through R_f is I_f . We assume that this is a ideal op-amp, so, no current is flowing in this op-amp. Due to negative feedback the node 'x' is virtually grounded and $V_x = 0$.

*Limitations of differentiator:

1. Highly sensitive to noise.
2. Affected by component tolerance.
3. Prone to D.C offset and drift.
4. Limited effective frequency range.

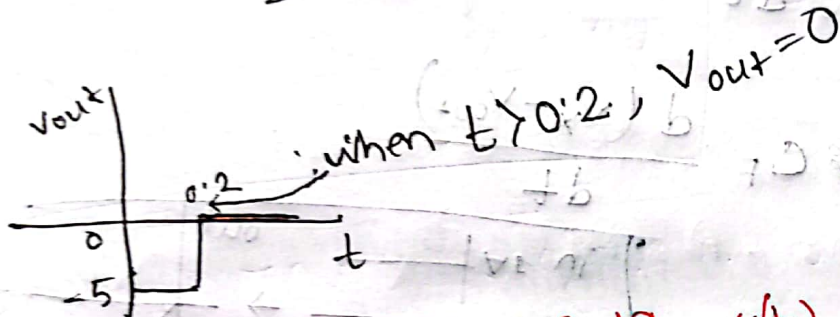
* Determine the output voltage:



Soln:
$$v_{out} = -R_f C \frac{dv_{in}}{dt}$$

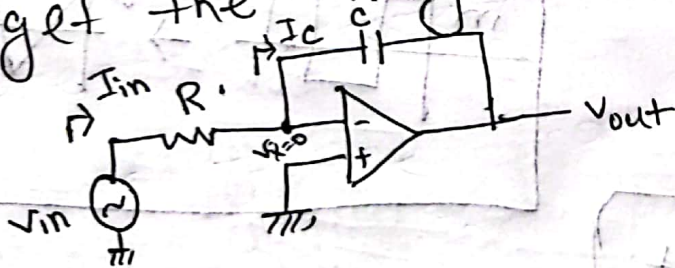
$$= -100 \times 10^3 \times 10 \times 10^{-6} \times \frac{1}{0.2}$$

$$= -5V$$



*** Op-Amp Integrator**

If we replace the feedback resistor of the inverting configuration with a capacitor, then we get the integrator circuit.



Let us consider the current flowing through R is I_{in} and current flowing through capacitor C_F is connected as the feedback. Because the feedback, the node x is virtually grounded.

So, at this node $V_x = 0$.

Applying KCL,

$$I_{in} - I_c = 0$$

$$\Rightarrow I_{in} = I_c$$

$$\Rightarrow \frac{V_{in} - V_x}{R} = C_F \frac{dV_c}{dt}$$

$$\frac{V_{in} - 0}{R} = C_F \frac{d(V_x - V_{out})}{dt}$$

$$\Rightarrow \frac{V_{in}}{R} = -C_F \frac{dV_{out}}{dt}$$

$$\Rightarrow \frac{d}{dt} V_{out} = -\frac{V_{in}}{R C_F}$$

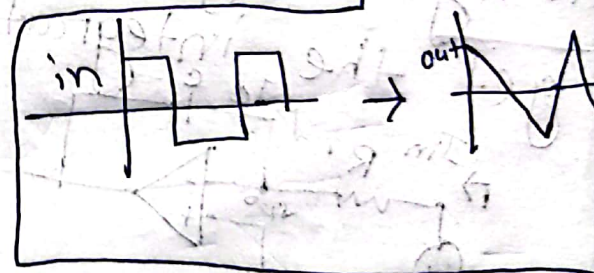
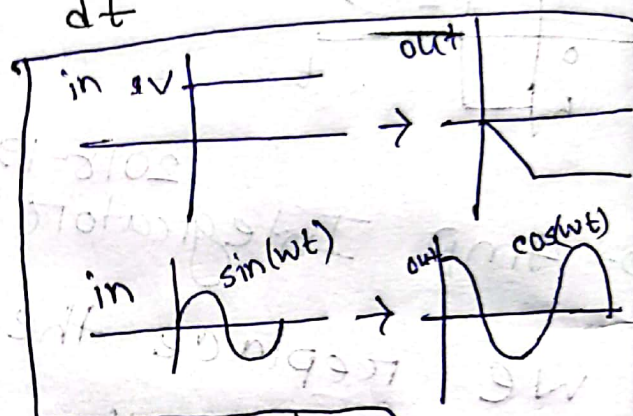
After Taking integration in both side,

$$V_{out} = -\int \frac{V_{in}}{R C_F} dt$$

$$V_{out} = -\frac{1}{R C_F} \int V_{in} dt$$

Here, by this equation, we get the integration of input voltage times the factor $\frac{1}{R C_F}$ as the output voltage.

V_c is voltage across capacitor
 $Q = C V$
 $\frac{dQ}{dt} = C \frac{dV_c}{dt} = I_c$



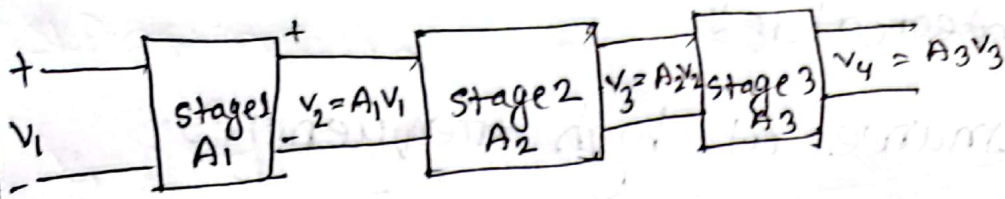
* Limitations of Integrators:

1. Affect performance at high frequencies.
2. Limits output when input is too large or long lasting.
3. Require careful handling of initial condition.
4. Amplifies noise, specially at low frequencies.

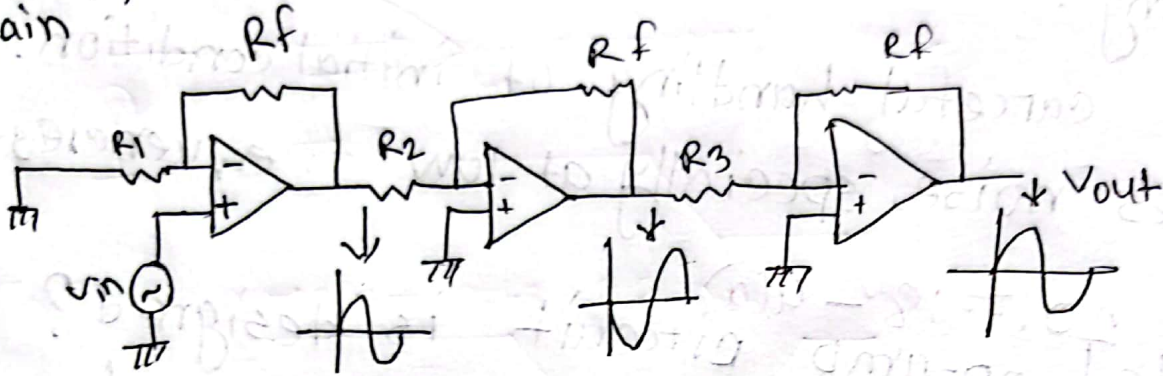
2017-18 - u(a)
How cascaded op-amp circuit is designed?
How output voltage is obtained in cascaded op-amp circuit?

Soln:
একটা op-amp circuit এর output ডারেকটা op-amp circuit এর input হিসেবে কাজ করে।

The head to tail arrangement of two or more op-Amp circuit is known as op-Amp cascaded connections. When op-amp circuits are cascaded, each circuit in the string is called stage.
Since the output of one stage is the input to the next stage, overall gain is the product of the gain of individual op amp circuit.



overall gain, $A = A_1 A_2 A_3$



As, op-amp stages are connected in series.

So, overall voltage gain, $A = A_1 A_2 A_3$

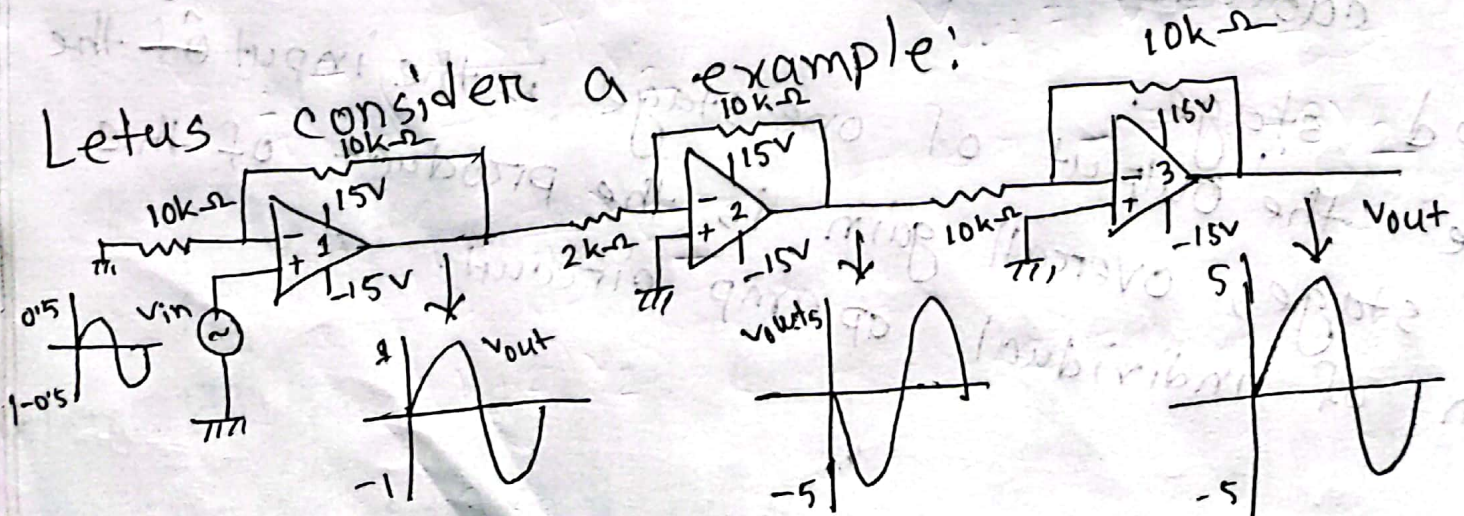
where, $A_1 = 1 + \frac{R_f}{R_1}$

$$A_2 = -\frac{R_f}{R_2}$$

$$A_3 = -\frac{R_f}{R_3}$$

indicating phase shift 180°

Let us consider an example:



In 1st stage,
It is non-inverting,

$$\text{So, } A_{v1} = 1 + \frac{R_f}{R_i}$$

$$= 1 + \frac{10}{10}$$

$$= 1 + 1 = 2$$

In 2nd stage,
It is inverting,

$$A_{v2} = -\frac{R_f}{R_2} = -\frac{10}{2} \text{ V} \times \left(\frac{1}{2}\right) = -5 \text{ V}$$

In 3rd stage,
It is inverting

$$A_{v3} = -\frac{R_f}{R_3} = -\frac{10}{10}$$

$$= -1$$

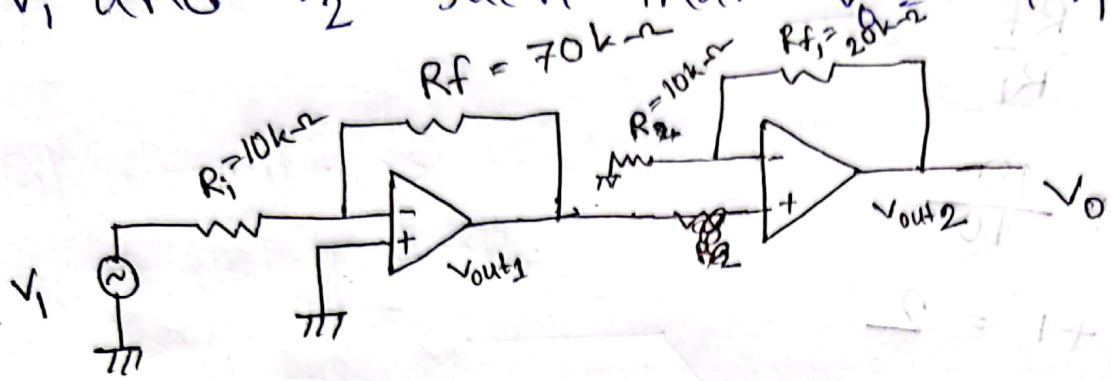
overall gain, $A = A_1 A_2 A_3 = 2 \times (-5) \times (-1)$

We know,

$$V_{out} = A \times V_{in} = 10 \times 0.5$$

$$= 5 \text{ V}$$

*Design an op-amp circuit with input V_1 and V_2 such that $V_{out} = -7V_1 + 3V_2$



In this circuit we get,

$$V_{out} = -\left(\frac{R_f}{R_i}\right) \times V_1$$

As given, The ratio needs to be 7 then

$$V_{out1} = -\left(\frac{70}{10}\right) V_1$$

$$= -7V_1$$

In 2nd circuit,

$$V_{out2} = \left(1 + \frac{R_f}{R_2}\right) V_2$$

$$= \left(1 + \frac{20}{10}\right) V_2$$

$$= (1 + 2) V_2$$

$$= 3V_2$$

$$V_{out} = V_1 + V_2$$

$$= -7V_1 + 3V_2$$